

RESEARCH ARTICLE | AUGUST 26 2024

Mining experimental magnetized liner inertial fusion data: Trends in stagnation morphology

Special Collection: [Papers from the 4th International Conference on Data-Driven Plasma Science](#)

William E. Lewis ; David A. Yager-Elorriaga ; Christopher A. Jennings; Jeffrey R. Fein ; Gabriel A. Shipley ; Andrew Porwitzky ; Thomas J. Awe ; Matthew R. Gomez ; Eric C. Harding ; Adam J. Harvey-Thompson ; Patrick F. Knapp ; Owen M. Mannion ; Daniel E. Ruiz ; Marc-Andre Schaeuble ; Stephen A. Slutz ; Matthew R. Weis ; Jeffrey Woolstrum ; David J. Ampleford ; Luke Shulenburg 



Phys. Plasmas 31, 082710 (2024)

<https://doi.org/10.1063/5.0206222>

 CHORUS



Articles You May Be Interested In

Integrated simulations of premagnetized and self-magnetizing dynamic screw pinch-driven MagLIF

Phys. Plasmas (May 2025)

Assessment of dynamic-screw-pinch-driven, current-scaled MagLIF target implosion performance using 3D magnetohydrodynamic simulations

Phys. Plasmas (October 2025)

Assessing the performance of MagLIF with 3D MHD simulations

Phys. Plasmas (February 2025)



Physics of Plasmas

Special Topics Open
for Submissions

[Learn More](#)



Mining experimental magnetized liner inertial fusion data: Trends in stagnation morphology

Cite as: Phys. Plasmas **31**, 082710 (2024); doi: [10.1063/5.0206222](https://doi.org/10.1063/5.0206222)

Submitted: 29 February 2024 · Accepted: 1 August 2024 ·

Published Online: 26 August 2024



View Online



Export Citation



CrossMark

William E. Lewis,^{1,a)} David A. Yager-Elorriaga,¹ Christopher A. Jennings,¹ Jeffrey R. Fein,¹ Gabriel A. Shipley,² Andrew Porwitzky,¹ Thomas J. Awe,¹ Matthew R. Gomez,¹ Eric C. Harding,¹ Adam J. Harvey-Thompson,¹ Patrick F. Knapp,² Owen M. Mannion,¹ Daniel E. Ruiz,¹ Marc-Andre Schaeuble,¹ Stephen A. Slutz,¹ Matthew R. Weis,¹ Jeffrey Woolstrum,¹ David J. Ampleford,¹ and Luke Shulenburg¹

AFFILIATIONS

¹Sandia National Laboratories, Albuquerque, New Mexico 87185, USA

²Los Alamos National Laboratory, Los Alamos, New Mexico 87545, USA

Note: This paper is part of the Special Collection: Papers from the 4th International Conference on Data-Driven Plasma Science.

^{a)}Author to whom correspondence should be addressed: willewi@sandia.gov

ABSTRACT

In magnetized liner inertial fusion (MagLIF), a cylindrical liner filled with fusion fuel is imploded with the goal of producing a one-dimensional plasma column at thermonuclear conditions. However, structures attributed to three-dimensional effects are observed in self-emission x-ray images. Despite this, the impact of many experimental inputs on the column morphology has not been characterized. We demonstrate the use of a linear regression analysis to explore correlations between morphology and a wide variety of experimental inputs across 57 MagLIF experiments. Results indicate the possibility of several unexplored effects. For example, we demonstrate that increasing the initial magnetic field correlates with improved stability. Although intuitively expected, this has never been quantitatively assessed in integrated MagLIF experiments. We also demonstrate that azimuthal drive asymmetries resulting from the geometry of the “current return can” appear to measurably impact the morphology. In conjunction with several counterintuitive null results, we expect the observed correlations will encourage further experimental, theoretical, and simulation-based studies. Finally, we note that the method used in this work is general and may be applied to explore not only correlations between input conditions and morphology but also with other experimentally measured quantities.

© 2024 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0206222>

I. INTRODUCTION

Magnetized liner inertial fusion (MagLIF)^{1,2} is a magneto-inertial fusion concept being studied at the Sandia National Laboratories Z-facility. Fusion-relevant conditions are achieved by compressing a premagnetized and preheated deuterium fuel contained in a cylindrical beryllium tube or liner with the magnetic pressure generated by a 16–20 MA current. External field coils³ premagnetize the fuel with an axially oriented magnetic field of 10–20 T, and a kilojoule-class laser preheats the fuel by coupling 0.7–2.4 kJ of energy.^{4,5} As the liner is accelerated inward due to the high magnetic pressure generated by the drive current, the fuel is quasi-adiabatically heated through mechanical PdV work. This also results in compression of the pre-imposed magnetic flux, which aids in the reduction of thermal conduction losses and trapping of charged fusion products. The implosion decelerates when the thermal pressure in the fuel becomes sufficiently high,

eventually resulting in stagnation with ion temperatures $T_{\text{ion}} \approx 3$ keV, fuel radial areal densities $(\rho R)_{\text{fuel}} \approx 2$ mg/cm², and several kiloTesla magnetic fields.^{6–8}

Ideally, the dynamics of this system depend only on a radial coordinate and a time coordinate as shown in Fig. 1. However, a variety of processes are hypothesized to seed growth of three-dimensional spatial structures that may lead to reduced performance through mixing of impurities into the fusion fuel as well as loss of plasma confinement⁹ and degradation of fuel pressure.¹⁰ For example, Fig. 2(a) shows two horizontal slices through a 3D tomographic reconstruction of the standard beryllium S65 material used for MagLIF liners. The image shows voids and inclusions of order several tens of micrometers in radius. When such defects are near the surface of the liner, they will seed the electro-thermal instability (ETI),^{11,12} which is one of the mechanisms thought to provide a seed for the magneto-Rayleigh–Taylor instability

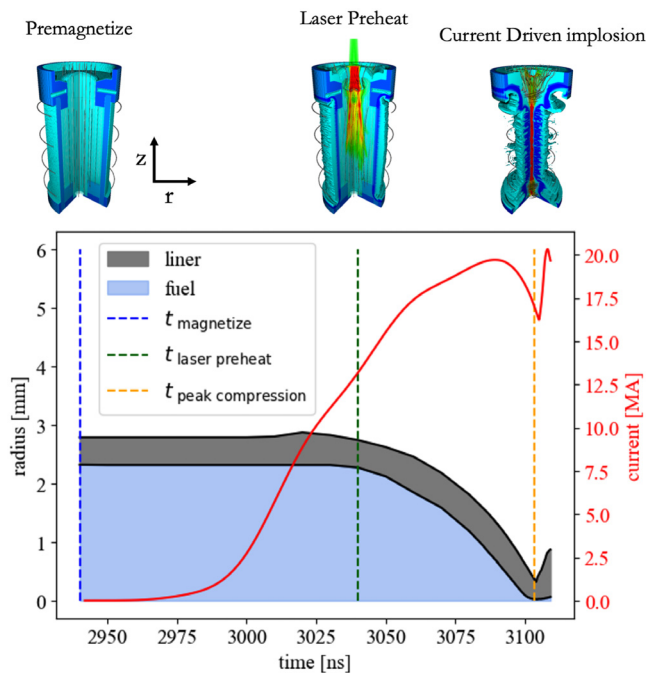


FIG. 1. The three key stages of MagLIF are axial premagnetization, laser preheat, and magnetically driven compression of a cylindrical beryllium liner containing deuterium fusion fuel. The included plot shows the trajectory of the inner and outer liner radii, as well as current vs time on the Z-machine timing basis. Ideally, the cylindrical geometry is preserved so that the dynamics are purely radial; however, a variety of experimental realities can lead to the development of three-dimensional structures, as discussed in the main text.

(MRTI) observed with in-flight radiography of MagLIF liners¹³ shown in Fig. 2(b).

The laser preheat stage also presents an opportunity for three-dimensional non-uniformities to develop. Since the first fully integrated MagLIF experiments published in 2014,⁶ a variety of different approaches to deliver laser energy to the fuel have been developed in offline surrogate gas-cell experiments. Generally, the goal of these efforts has been to improve energy coupling and/or allow for increased initial fuel pressure while avoiding increased mix resulting from thicker laser entrance foils typically needed to contain the gas at higher fill pressures.^{15–21} Regardless of the approach, the laser penetrates the laser entrance window (LEW), allowing the fuel to escape through the LEW, as indicated in interferometric imaging of offline experiments shown in Fig. 2(c).²⁰ In the experiment shown in Fig. 2(c), the target does not implode. As a result, the timescale for gas to escape is on the order of several microseconds, much longer than the roughly 60 ns it takes after the laser preheat for a MagLIF target to implode. In integrated MagLIF experiments, end losses are still expected from both the laser entrance hole and the gas fill inlet at both the top and bottom of the target as a result of compression. This may result in large-scale axial non-uniformities of stagnation conditions. In addition, Fig. 2(d) clearly demonstrates that delivering laser preheat from the top side of the target introduces axial non-uniformities in energy deposition,^{18,20} as well as LEW mix.¹⁷ It further illustrates that the beam exhibits asymmetric filamentation as it propagates into the fuel.¹⁸ This is

consistent with behavior in HYDRA simulations,²² and leads to non-uniform deposition of energy in the fuel. HYDRA simulations also predict that this process introduces significant vorticity into the imploding fuel column that may contribute to the development of radial and azimuthal structure in the fuel plasma.

Other aspects of MagLIF target design impact the extent to which the various seeds for three-dimensional structure described above are expected to play a role. For example, the extent to which MRTI feeds through the liner is expected to depend on the liner acceleration history and aspect ratio,²³ given by $AR = R_{0,out}/(R_{0,out} - R_{0,in})$, where $R_{0,out(in)}$ is the initial outer (inner) liner radius. This expectation has been shown to be consistent with small targeted experimental datasets.^{24,25} Reference 25 also showed by analyzing a small number of experiments that dielectric coatings, which are expected to mitigate ETI growth,^{9,26,27} appear to reduce the amplitude of helical structure in self-emission images of the stagnated fuel plasma and improve stagnation performance reproducibility.²⁵ This is consistent with the hypothesis that coatings reduce the seed for and therefore growth and feedthrough of the MRTI, which can degrade plasma confinement and reduce fusion yield.⁹

From the diagnostic standpoint, configuration details of the spherical crystal imager¹⁴ used to observe 2D projections of the self-emission may also be important. For example, two imagers can be fielded along multiple lines of sight. These may show distinct morphology and allow for an approximate tomographic reconstruction.²⁸ The spatial resolution, magnification, and spectral sensitivity can also depend on the configuration. These configuration choices have the potential to impact *apparent* structure and should be accounted for when trying to make fair comparisons between experiments. To indicate the types of structures observed, representative stagnation images across a range of different experimental input conditions and spherical crystal imaging modalities are shown in Fig. 3. These images demonstrate rich morphological variety that is typical of our entire MagLIF image database.

The current paradigm for understanding observed structure in MagLIF stagnation images is to hypothesize what physics is likely to dominate the formation of particular structures in the morphology. We then perform a small experimental study or mine our data for “directly comparable” experiments that significantly change only one experimental input feature corresponding to the expected dominant physics. To date, this approach has resulted in significant gains in understanding and improved performance of MagLIF experiments.^{7,27} However, there are many aspects of the experiment design and hardware that have changed over time, frustrating prospects for explicit A/B comparisons to assess the impact of specific input changes on three-dimensional structures. It is therefore feasible that previously unexplored mechanisms provide a seed for or may help to mitigate development of three-dimensional structure.

In this manuscript, we demonstrate the use of a Bayesian statistical regression framework to test for linear correlations between experimental conditions and MagLIF stagnation image structure. The framework allows for any number of experimental input features to be incorporated, including slotted current return can structure, which was not previously thought to significantly impact the development of three-dimensional structures. We emphasize that the regression-based approach is completely general and could be applied to analyze correlations between experiment configuration and information from other

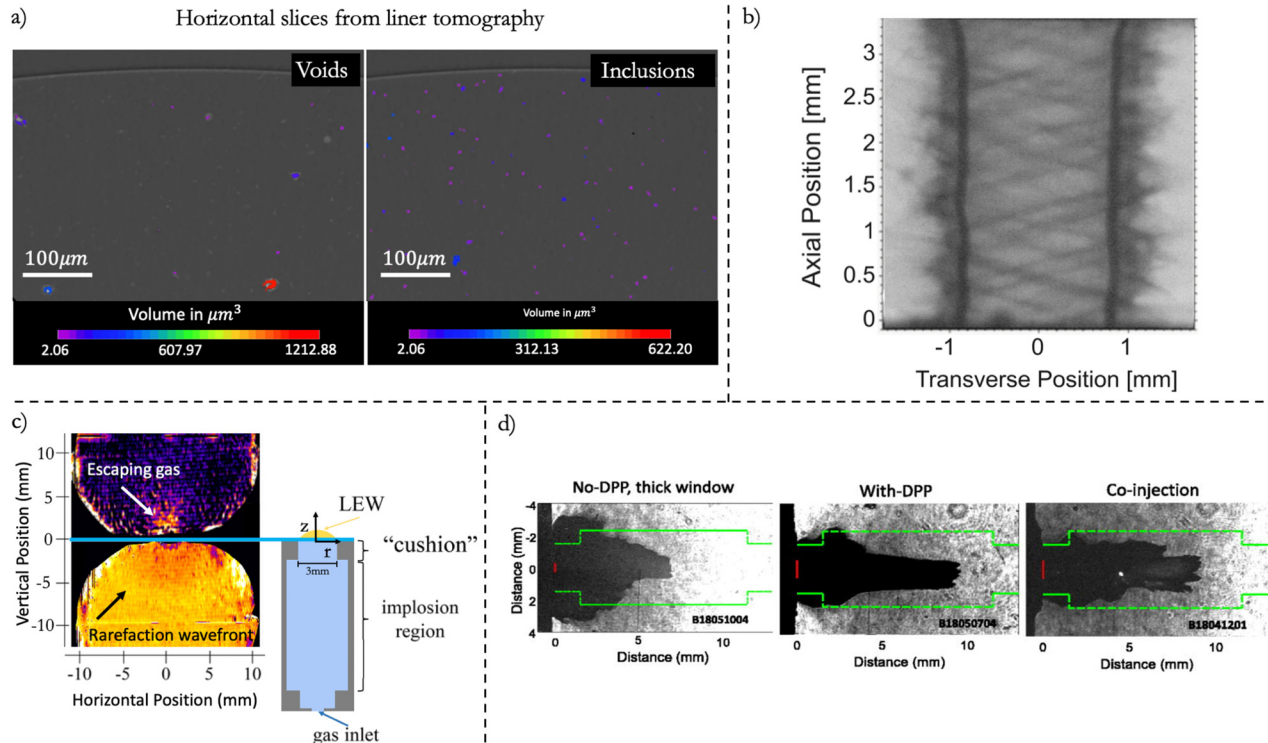


FIG. 2. Various experimental realities exist that spoil the ideal purely radial implosion structure in MagLIF experiments. (a) Tomography of the standard beryllium S65 material used for MagLIF liners shows the existence of inclusions of several tens of micrometers in radius. Voids and inclusions near the outer surface of the liner give rise to growth of the electro-thermal instability (ETI). (b) In-flight radiographs of the helical magneto-Rayleigh Taylor instability (MRTI) developed on a MagLIF liner. This is hypothesized to be seeded by ETI and/or Hall physics and low-density power-feed plasma. The instability is expected to feedthrough the liner to an extent dependent on the liner thickness. This will imprint some structure onto the stagnation, degrading compression and confinement. Adapted with permission from Awe *et al.*, *Phys. Rev. Lett.* **111**, 235005 (2013). Copyright 2013 American Physical Society. (c) Interferometric imaging of the gas dynamics during laser preheat. The laser penetrates the laser entrance window (LEW), leaving an opening through which the gaseous fill can escape. In integrated MagLIF experiments, losses from the LEW and gas inlet at the bottom of the target occur primarily during the compression phase. This may result in some large-scale axial variation in stagnation conditions. Adapted with permission from Galloway *et al.*, *Phys. Plasmas* **28**, 112703 (2021). Copyright 2021 AIP Publishing LLC. (d) Shadowgraphs of the blast wave initiated by the laser preheat in offline gas-cell experiments. Variation in preheat “protocols” shown in this figure include differences in LEW thickness, whether or not distributed phase-plate (DPP) beam smoothing was applied, and the use of a foot-pulse reducing LEW mix (co-injection). Note that the different protocols result in dramatically different structures both axially and radially. These differences may impact the development of three-dimensional structure during the implosion. Adapted with permission from Harvey-Thompson *et al.* *Phys. Plasmas* **26**, 032707 (2019). Copyright 2019 AIP Publishing LLC.

diagnostics. The particular problem being addressed, however, will in general require careful curation of data for consideration.

This work is organized as follows: In Sec. II, we discuss the creation of a structured dataset containing detailed information about experimental configurations and image metrics from a variety of structured and unstructured data sources. These data sources include hardware drawings, text data, and image plate scans. In Sec. III, we present the Bayesian regression framework, describing how the results will be interpreted. In Sec. IV, we apply the methodology to our dataset. We find our data supports several previously discussed effects, providing some validation of the approach. We also highlight a number of new mechanisms that appear to impact stagnation morphology. We conclude in Sec. V by providing a summary of key findings as well as a discussion of areas where our methodology may be applied to develop new insights.

II. EXPERIMENTAL DATASET STRUCTURE

MagLIF data sources include “structured” data (i.e., numerical and categorical data contained in a standardized data format) from a

variety of diagnostics, as well as significant amounts of “unstructured” data (e.g., hardware drawings, diagnostic configuration summaries, and raw image data) requiring preprocessing for use in a structured form. In this section, we document our approach to mining the data from these sources, including the analysis methods used. However, we note that many of these analyses have been documented elsewhere and have since become routine. For conciseness, where appropriate, we cite existing literature and utilize the resulting structured dataset, reserving significant detail only for those methods, which have not been previously discussed in the literature. To simplify the presentation, we split the discussion into two subsections, Secs. II A and II B, describing the input and output data, respectively.

We note that, due to the high-dimensional unstructured nature of our initial dataset, some judgment is made regarding what data are reasonable to include and how to process such information into a usable form. While we are purposefully broad when selecting input data to include, we do not attempt to be completely exhaustive. For example, we do not include information regarding geometry of the

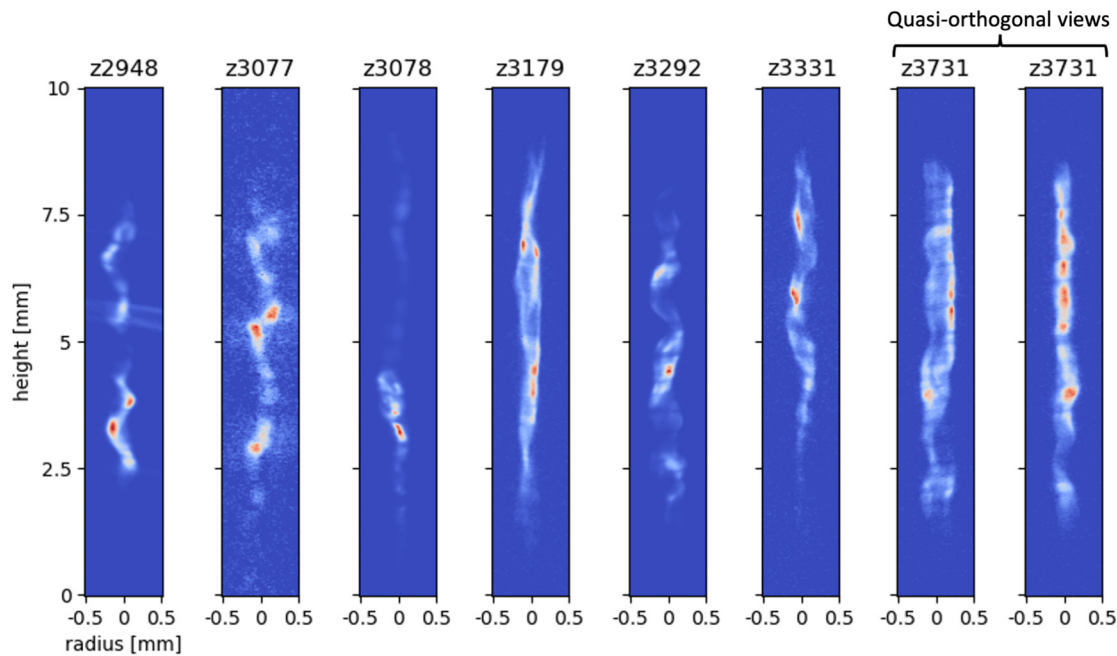


FIG. 3. MagLIF stagnation images show rich variety in their structure. The images above are representative of the wide range of experimental hardware, target geometry, and spherical crystal imaging modalities. The data illustrate that apparent structure, or lack thereof, may be due to the combination of these different factors. For example, a lower resolution imager (as used in z2948 and z3077) may leave radial structure unresolved. Imaging a combination of continuum and a cobalt He- α spectral line, as in the case of z3078, results in brighter emission at the bottom of the strand due to the lower part of the inner target wall being coated in cobalt.¹⁴ Changing target or hardware configurations (all shot numbers) or quasi-orthogonal views (z3731) also results in varied structure.

Helmholtz coils used to premagnetize the fuel. In developing coils to achieve higher magnetic field strength, this geometry has changed.²⁷ However, it is reasonable to presume that, to first order, it is the magnetic field strength rather than coil geometry that is more likely to impact stagnation morphology. Regardless, it is easy to see how the method extends to new data sources. Furthermore, our dataset includes experimental factors that have not been previously discussed in the literature, and we find evidence consistent with previously unconsidered causal effects discussed in Sec. IV. Our findings lay the foundation for detailed exploration in future work. Finally, we note that our data consist only of “integrated” MagLIF shots that included all four features of a gaseous deuterium fuel fill, external axial magnetic field, laser preheat, and current-driven compression, and which produced x-ray stagnation images that could be analyzed for image metrics. In some cases, e.g., where the image plate recording the data was too heavily damaged by debris, image metrics could not be reliably computed. The result is a dataset including a total of 57 experiments, seven of which included two quasi-orthogonal views of the stagnation as described in Sec. II B.

A. Inputs: Experiment configuration

To help organize our discussion, we work with several conceptual sub-categories detailed in Secs. II A 1–II A 4. Throughout, we will use square brackets, for example, $\xi \in [a, b]$, to indicate variables ξ that are fundamentally continuous on the interval a to b . However, we note that in many cases, only a small number of discrete values are fielded experimentally for the continuous parameters. For quantities that are

categorical in nature, we use curly brackets $\xi \in \{a, b\}$, listing the set of values used.

1. Liner parameters

MagLIF liners fielded experimentally have the following features, where we include the current range of parameter values fielded on Z:

- Initial inner liner radius $R_{0,\text{in}} \in [2.04, 2.325]$ mm.
- Initial outer liner radius $R_{0,\text{out}} \in [2.61, 2.97]$ mm.
- Liner height $h \in [7.5, 10]$ mm.
- Liner coating (none or mass-matched dielectric coating defined in Ref. 25) with indicator variable $C \in \{0, 1\}$.

Figure 4 shows two examples of MagLIF liners indicating these variables in each case. In all cases, indicator variables set to 1 indicate that the condition was applied (e.g., $C=1$ indicates presence of a mass-matched dielectric coating).

Also note that Fig. 2(a) indicates that we are beginning to collect detailed information on the structure of voids and inclusions in MagLIF liners. The statistics of these voids and inclusions are expected to depend on a variety of factors, such as material batch and the liner metal alloy, which include the standard beryllium S65, high-purity beryllium, or the presence of cobalt coatings intended to track the presence and location of mix.¹⁴ While we do not currently have sufficient liner tomography data to make use of this information in our current analysis, correlating the distribution of void and inclusion sizes to stagnation morphology and performance may be viable in the relatively

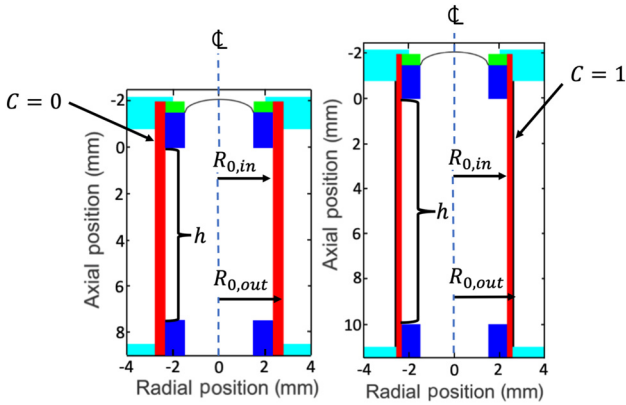


FIG. 4. Two examples of vertical slices through MagLIF targets indicating the liner parameters discussed in Sec. II A 1 are shown. In particular, inner and outer initial liner radii $R_{0,in(out)}$, the height of the imploding region of the liner h , and a variable C indicating whether or not a mass-matched dielectric coating is applied are used to describe the liner in our regression analysis. For the target on the right, the coating is indicated by the black region on the outside wall of the liner shown in red. We refer the interested reader to Ref. 17 for a discussion of additional target features not relevant to this work.

near future. We expect that our regression approach may prove useful for gaining detailed insights when more data becomes available.

2. Integration parameters

Integrated MagLIF shots require the liner to be filled with a pre-magnetized fusion fuel (DD or DT). Standard targets use a gaseous DD fuel at a specified mass density (or pressure), although mixtures of DD and DT as well as ice layers are in development. The preheat laser couples energy into the fuel to set the initial adiabat, with subsequent quasi-adiabatic compression from the current pulse. The rise time of the current is approximately 100 ns, with peak current determined by the Z-machine charge voltage and the inductance history of the target. These integration parameters are given by

- Gas fill density $\rho_{fill} \in [0.7, 1.4]$ mg/cc.
- Laser energy deposited $E_{preheat} \in [0.7, 2.4]$ kJ.
- Applied axial magnetic field $B_{z,0} \in [10, 20]$ T.
- Peak current $I_{max} \in [14, 21]$ MA.

A diagram indicating these parameters is provided in Fig. 5. As mentioned above, the peak current and exact details of the current rise depend on the inductance history. This is generally determined by a large number of parameters such as the machine charge voltage, initial inductance of the target, and current return “can,” as well as the liner inductance history, set to first order by the radial and axial dimensions of the liner. It may be possible in general, to perform our analysis using initial inductance computed from target and power-feed geometries, some of which are detailed in Ref. 27. However, we believe that utilizing peak current determined from velocimetry where available, and by imputation from comparison to similar targets when it is not available, will prove more physically intuitive. We also note that work has been done to explore the application of pulse shaping for quasi-isentropic compression of MagLIF liners.^{29,30} If pulse shaping begins to be applied to integrated MagLIF implosions, it may become necessary to

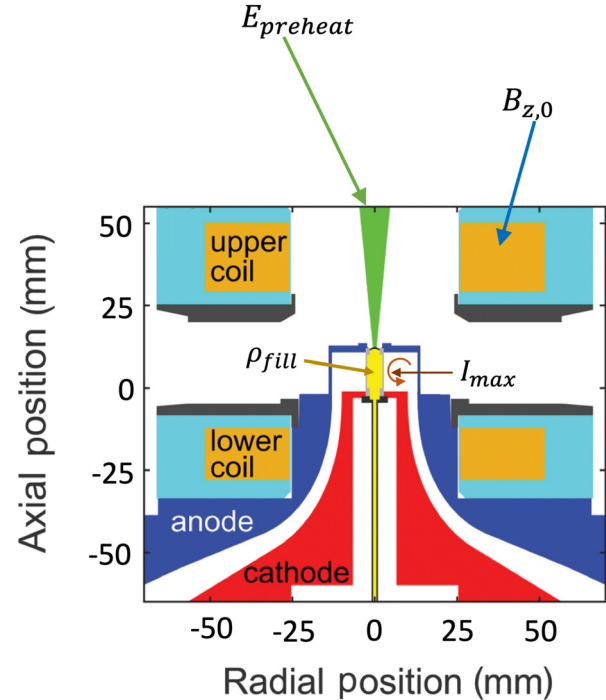


FIG. 5. An example of a MagLIF load including power feed and magnetic field coils is shown. We indicate the four MagLIF integration parameters discussed in Sec. II A 2 that appear in our regression analysis.

incorporate more detailed current drive information into future analyses.

3. Preheat and magnetic drive protocol parameters

For this group of parameters, we include a characterization of details relating to how the laser preheat^{15–18,21} and electrical energy³¹ from Z are coupled to the target:

- Distributed phase-plate conditioning of the preheat laser beam with indicator variable $DPP \in \{0, 1\}$.
- Thickness of the LEW $\Delta_{LEW} \in [0.5, 3.5]$ μm .
- Azimuthal drive asymmetry amplitudes $C_{m,a(r)} \in [0, C_m]$ (dependent on return can structure and mode number discussed in the text below).
- Azimuthal drive asymmetry view factors $c_{m,a(r)}(\phi) \in [-1, 1]$ (dependent on return can structure, mode number, and line of sight (LOS) discussed in the main text below).

The first two parameters relate to how laser energy is delivered to the target and are indicated in Fig. 6. We note that we do not include the presence of cryogenic cooling in our regression, which is what enables LEWs with a thickness of $0.5 \mu\text{m}$.²¹ This feature is not included as we expect the beam conditioning, which modifies laser filamentation, and LEW thickness, which modifies beam propagation depth,¹⁸ are likely to be more dominant in impacting stagnation morphology. We also note that Δ_{LEW} is reported as the nominal thickness of the window prior to deformation from filling the target to the initial fill density.

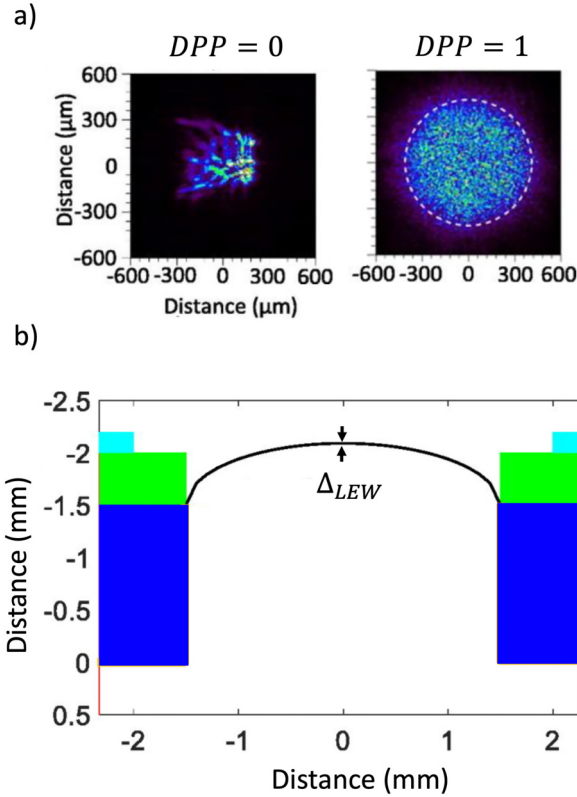


FIG. 6. (a) Distributed phase-plate conditioning of the preheat laser profile results in a significantly more uniform beam profile. Adapted with permission from Harvey-Thompson *et al.*, Phys. Plasmas **26**, 032707 (2019). Copyright 2019 AIP Publishing LLC. (b) Vertical cutaway of the top portion of a MagLIF target showing the LEW. Depending on the gas fill pressure and choices in how laser energy is delivered, different thicknesses of the LEW foil Δ_{LEW} can be fielded.

Figure 7 demonstrates the steps used to characterize the overall amplitudes of any drive asymmetry in terms of azimuthal harmonics with mode number $m = 0, 1, 2, 3, \dots$. First, hardware drawings including the return can structure with slots [e.g., Fig. 7(a)] are used to define the return can structure in an electromagnetic field solver developed following the method described in the appendix of Ref. 32. The field strength for the return can in Fig. 7(a) is shown in the $r - \phi$ plane in Fig. 7(b). We then extract the radial and azimuthal magnetic field components at a radius of $R_{drive} = 3.36$ mm, sufficiently large to lie outside of any of our liner outer radii. Figure 7(c) shows the radial (left) and azimuthal (right) field components vs ϕ . To obtain mode amplitudes, we perform a Fourier analysis of the azimuthal and radial magnetic field components at the specified radius of 3.36 mm separately,

$$B_{a(r)}(\phi) = \sum_{m=0}^{\infty} a_{m,a(r)} \cos(m\phi) + b_{m,a(r)} \sin(m\phi), \quad (1)$$

where $B_{a(r)}$ is the azimuthal (radial) magnetic field component as a function of ϕ , m is the azimuthal mode number, and $a_m(b_m)$ are the amplitudes of the cosine and sine terms in the decomposition. We may then obtain mode amplitudes

$$c_{m,a(r)} = \sqrt{a_{m,a(r)}^2 + b_{m,a(r)}^2}. \quad (2)$$

The extracted mode amplitudes for the return can structure in Fig. 7, which are given in panel d. Note that the $m = 0$ mode vanishes identically for all radial drive fields due to $\nabla \cdot \mathbf{B} = 0$. The magnetic field is also normalized so that the line integral around the circle centered at $r = 0$ mm and containing the central conductor gives 2π . Note that this normalization is independent of the current used in the field solver so that all of the drive asymmetries are normalized in the same fashion. In doing so, we implicitly assume that the drive asymmetry is independent of drive current and other environmental factors.

The last set of parameters included in this grouping is what we will refer to as mode-number-dependent ($m = 1, 2, 3, \dots$) asymmetry view factors $c_{m,a(r)}(\phi)$ that depend on LOS specified by view angle ϕ and field component a or r for azimuthal and radial, respectively. The reason for including this information is that our experiments view the stagnation along a particular LOS specified by the angle ϕ . It seems possible that the morphology may depend on the relative orientation between any drive asymmetry and the LOS. We therefore calculate the normalized mode amplitude (including sign) along each experimental LOS as

$$c_{m,a(r)}(\phi) = \frac{a_{m,a(r)} \cos(m\phi) + b_{m,a(r)} \sin(m\phi)}{c_{m,a(r)}}. \quad (3)$$

Figure 8 shows how this geometric quantity corresponds with two views for the $m = 2$ contribution to $B_a(\phi)$ for the return can structure in Fig. 7. View 1 gives $c_{2,a}(\phi_1) \approx -1$ and view 2 gives $c_{2,a}(\phi_2) \approx 0.5$. Note that for return cans with high symmetry, the mode amplitudes for the low modes will be identically zero due to symmetry arguments. In this case, there are two reasonable choices for the view factors, setting them to zero, or assigning a random value between $[-1, 1]$. We have chosen the former, but neither choice will strongly impact the regression analysis described in Sec. III since the average value of this input variable across the data with no low-mode drive asymmetry is the same in both cases. Finally, we note that here we discussed the presence of drive asymmetry in the radial magnetic field component for completeness. However, by comparing the left to right panels in Figs. 7(c) and 7(d), we note that the amplitude of the drive asymmetry in the radial magnetic field component is an order of magnitude smaller than for the azimuthal magnetic field component. Including information on drive asymmetry of $B_r(\phi)$ significantly increases the input space dimensionality. Furthermore, we did not observe significant correlations with morphology in our data. For these reasons, we shall not discuss this aspect further.

4. Diagnostic parameters

The various spherical crystal imaging configurations fielded on Z vary in both spatial and spectral response characteristics.¹⁴ As we will describe in Sec. II B, our image metrics are purposefully chosen to emphasize structural characteristics of stagnation morphology that are intended to be independent of brightness variations to the greatest possible extent. With this in mind, we do not consider spectral characteristics of the imager configurations in this work. This is assured by considering imagers that collect continuum self-emission. However, we note that depth-of-field, magnification, and point spread functions (PSFs) depend on configuration and also vary across the object plane,

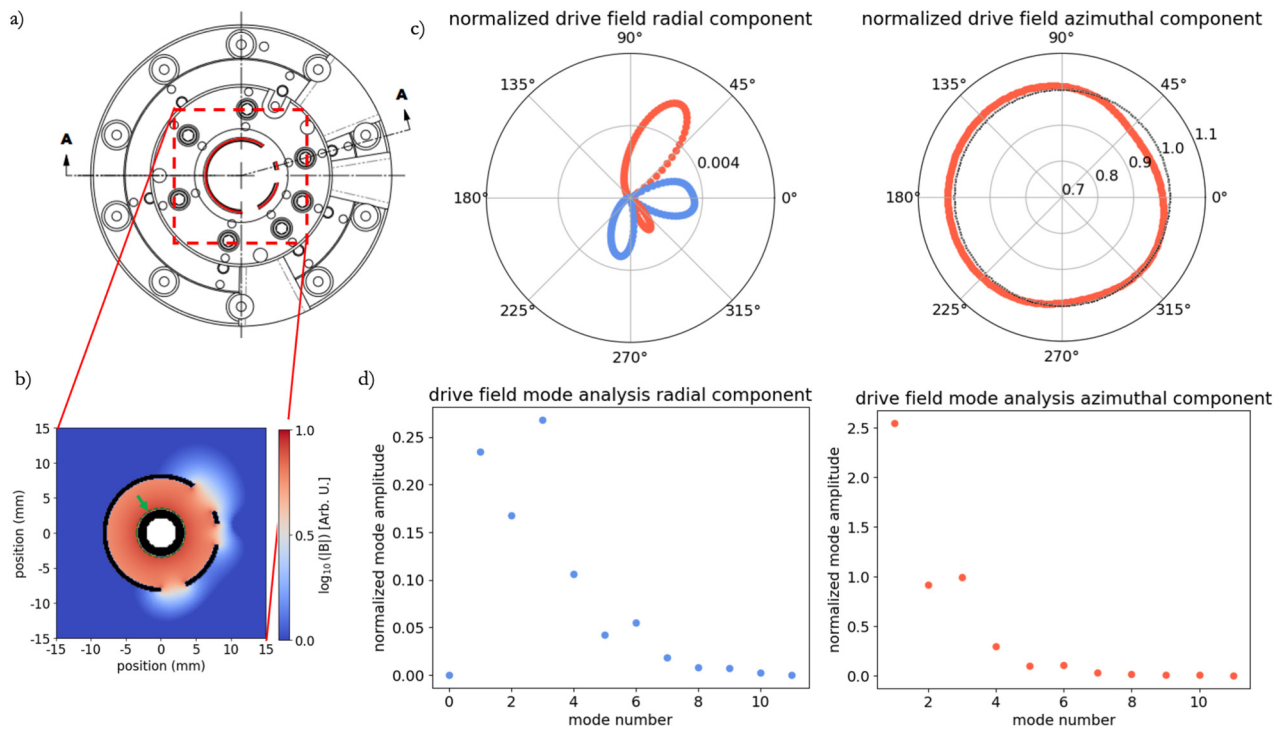


FIG. 7. (a) Load hardware drawing showing the slotted return can structure in red. (b) The hardware drawing is mapped to boundary conditions for an electromagnetic field solver to obtain drive asymmetry. The green arrow points to a dashed circle at a radius of 3.36 mm. This is just outside of the inner conductor shown in black and is the radius at which the fields are extracted for asymmetry analysis. (c) Radial (left) and azimuthal (right) field components can be extracted at a fixed radius (here 3.36 mm) for further analysis (blue indicates field points in the $-\hat{r}$ direction). (d) By performing a Fourier decomposition, drive field mode amplitudes may be extracted to characterize drive asymmetries.

and may impact apparent morphology. We work in the object plane when computing image metrics so that magnification is taken into account. We therefore expect that resolution will be the dominant imager characteristic contributing to observed structure according to the image metrics laid out in Sec. II B. For nominal imager configurations, we took values from Ref. 14 for resolution and magnification factors. In a small number of cases, the image plate was not fielded in the nominal location, and values were obtained for those cases separately.³³ The main parameters characterizing the imager in our regression analysis are the following:

- Horizontal imager resolution $H_{\text{res}} \in [15, 79] \mu\text{m}$.
- Vertical imager resolution $V_{\text{res}} \in [16, 120] \mu\text{m}$.

We also note that the different imager configurations exhibit different amounts of pincushion distortion across the object plane as indicated by Fig. 9. We do not attempt to quantify this as an input parameter here, as we expect resolution at the center of the object plane to be the diagnostic parameter that will have the greatest impact on our image metrics.

B. Outputs: Stagnation image morphology metrics

For our purpose, the rich structure in MagLIF stagnation images will be processed down to a small number of intuitive scalar quantities that can then be correlated with the experimental inputs described in

Sec. II A. There are several properties that we desire these metrics to obey. First, as we do not currently have absolutely or even relatively calibrated image plate intensities across the entire dataset, we consider metrics that are approximately independent of the overall scaling of the image intensity. While we will not explicitly prove this, the construction of our processing algorithms involves computations that are invariant to an overall multiplicative factor on the image intensity,^{25,34} so we expect our results to be independent of data calibration. Second, we will attempt to utilize metrics that characterize the geometric structural properties independently of axial brightness variations and spectral content. One reason for this is that spectrally dependent diagnostic response and axially varying liner absorption arising from undiagnosed variations in liner areal density will be more challenging to interpret. In particular, these factors will impact the amount of emission collected from different parts of the fuel plasma. However, Ref. 28 provides an argument that $\rho R(z, \phi)$ varies sufficiently slowly as a function of azimuthal angle ϕ so that, over the width of the stagnation column, we expect to be able to approximate $\rho R(z, \phi) \approx \rho R(z)$. The attenuation for a given slice will then not vary significantly across a given line-out. As a result, areal density effects should not strongly impact the estimation of the stagnation column boundary locations and width per our methods described in the next paragraph.

The metrics that we use are dimensionless scalar values computed from three different contours derived from a given image. These contours are a left and right boundary of the strand emission vs height as

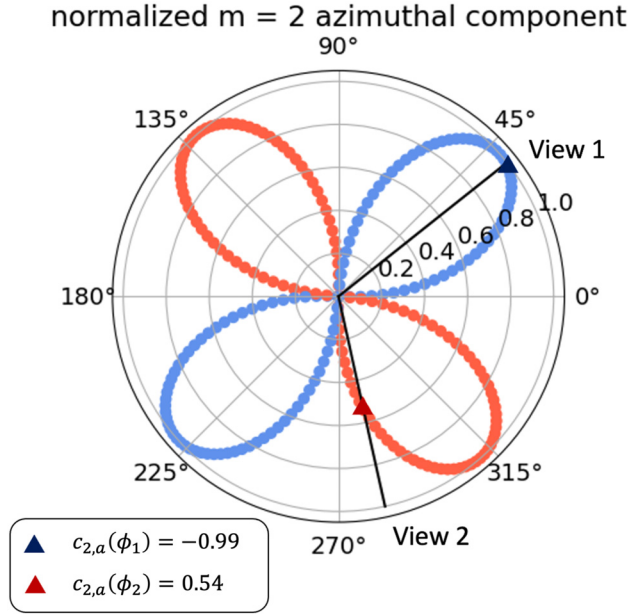


FIG. 8. The $m=2$ component of the azimuthal drive field shown in Fig. 7(c) is normalized to have an amplitude of unity. Note that the blue lobes point in the $-\phi$ direction and reduce the magnitude of the azimuthal drive, while the red lobes point in the $+\phi$ direction, increasing the drive field. As a result, one might expect development of asymmetries in the stagnated fuel column as discussed in more detail in Sec. IV C 2. Also note that, in general, the view angle of the spherical crystal x-ray imaging diagnostic relative to the return current can be known and can be used to calculate view factors for each mode number m as described in the text. For the return current can shown in Fig. 7(a), two quasi-orthogonal views are fielded as shown by the solid black radial lines. The view factor values along these lines are shown in the lower left legend.

well as the average position computed as the mean between these two boundaries. Examples of these contours are shown for three different stagnation images in Fig. 10. At each height, these boundaries are found using a fixed threshold intensity defined by $t * \mathcal{I}_{\max}(z)$, where $0 < t < 1$ is the threshold and $\mathcal{I}_{\max}(z)$ is the maximum intensity at a particular height. First, we perform polynomial background subtraction using off-strand background data to fit the polynomial at a given axial position. Subsequently, the contour boundaries are found by moving outward left or right from the point of peak intensity at a given axial position, respectively, and selecting the first pixel falling below the threshold intensity value at that axial height. For this work, we set a threshold of $t = 0.05$. This value was the lowest value considered, as lower values were found to be sensitive to image noise in determining the boundary, while increasing the threshold led to undesirable sensitivity of the metrics to “internal” brightness structure of the strand. The [supplementary material](#) includes a figure demonstrating the reasoning behind the choice of $t = 0.05$ in more detail. All images were checked by hand for quality. Only two out of a total of 64 experimental images required special treatment by hand, with details included in the [supplementary material](#). Finally, the supplement demonstrates that although variance is increased when utilizing a higher threshold or an inward rather than outward moving algorithm, there is a strong linear correlation, so that we did not find the conclusions of our manuscript

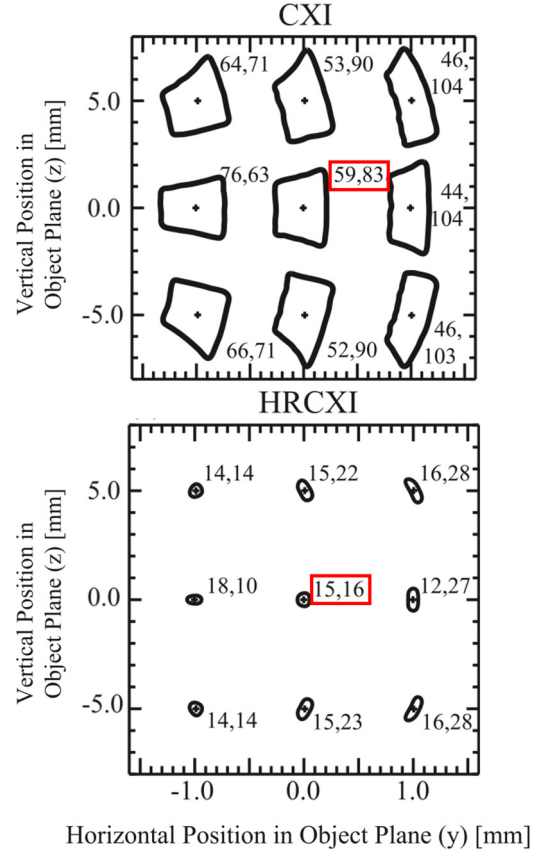


FIG. 9. Point spread functions in the object plane for two different spherical crystal imager configurations. We characterize the imager by the horizontal and vertical resolution near the center of the object plane (red box on each plot). Adapted with permission from Harding *et al.*, Rev. Sci. Instrum. **94**, 083509 (2023). Copyright 2023 AIP Publishing LLC.

to be impacted strongly by the choice of threshold and contour finding algorithm.

Our first image metric is defined by integrating the power spectrum computed from the mean position of the stagnation column normalized by the column height

$$E_s = \frac{1}{\mathcal{S}} \int_0^\infty |f(k)|^2 dk, \quad (4)$$

where

$$f(k) = \int_{\mathcal{S}} s(z) e^{-ikz} dz \quad (5)$$

is the Fourier transform computed by integrating over the strand height $\mathcal{S}$ of the mean strand position given by

$$s(z) = (s_r(z) + s_\ell(z))/2. \quad (6)$$

The strand position is a function of axial height computed from the right (left) strand boundary $s_{r(\ell)}(z)$ at height z . We note that, functionally, E_s is computed using the fast Fourier transform divided by the

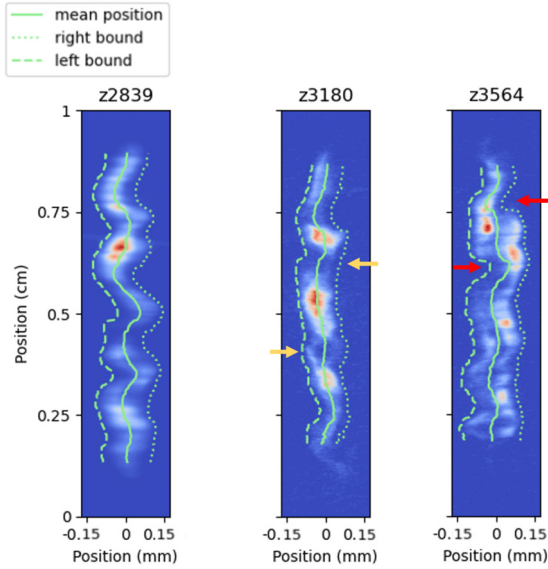


FIG. 10. Left and right strand contours, as well as the average between the two, are used as described in the text to compute several image metrics that characterize the geometric structure of MagLIF stagnation columns. Here, we show three examples, demonstrating that the structure is often able to be extracted relatively independently of axial brightness variations, even when helically entwined bifurcations are present (yellow arrows in the middle image). The third panel is an extreme case, where there appear to be two adjacent bifurcated strands with regions where one strand is significantly dimmer or apparently not present (red arrows). In such cases, it is not clear if this structure is associated with fuel plasma structure or liner attenuation. The resulting metrics may be somewhat biased by liner effects, but are expected on average to contain useful geometric information about the fuel plasma.

sampling frequency as detailed in Ref. 25. The metric E_s provides a means to characterize the total variation or amplitude of strand oscillation away from a straight vertical line and is therefore a measure of the overall kink-like structure. While we will not use more detailed information here, we note that access to the spectrum $|f(k)|$ may allow for more finely detailed metrics characterizing the scales present in the strand and may be interesting to investigate as more MagLIF experiments are performed. Additionally, we note the importance of the normalizing factor $1/S$. By Parseval's theorem, the integral of the power spectrum in Eq. (4) is proportional to S , the height of the strand. To characterize an amplitude of oscillation, we construct our metric to be independent of strand height by dividing out this dependency.

To compute the other two metrics, we define the axially resolved convergence ratio

$$CR(z) = \frac{R_{0,\text{in}}}{(s_r(z) - s_\ell(z))/2}. \quad (7)$$

Our second image metric is computed as the mean of the convergence ratio across the height of the stagnation column

$$\overline{CR} = \frac{1}{S} \int_S CR(z) dz, \quad (8)$$

which measures the average of the ratio of the initial inner liner radius to the radius of the strand. The third metric is defined as the standard deviation of the convergence ratio:

$$\sigma CR = \sqrt{\frac{1}{S} \int_S (CR(z) - \overline{CR})^2 dz}, \quad (9)$$

which measures the variation of the convergence ratio along the column height. By assessing variation in convergence ratio, σCR quantifies sausage-like structure in the column. However, some stagnation images exhibit entwined helical strands. It is possible that such structure will also contribute to increased apparent variation in CR across the stagnation column height, even if the overall cross-sectional area is stable. When interpreting the results of our regression analysis, we will refer to this metric as characterizing sausage-like structure, but the possibility of other morphological features contributing to this metric should be kept in mind.

While it is not possible to ensure that these metrics are completely insensitive to variations in liner areal density, we reemphasize that they are chosen to minimize the impact of this to the extent possible. Developing metrics that quantify the impact of liner attenuation and spectral line emission from impurities is beyond the scope of this work.

III. REGRESSION ANALYSIS

To analyze correlations in our dataset, we utilize Bayesian linear regression. A forward model is specified as $Y = AX + b + \epsilon$. Here, Y is a vector of output values for each experiment, X is the matrix of input features where each row is a particular experiment and each column contains a particular experiment input feature, and A and b are a vector and scalar containing free parameters to be fit. The error term, $\epsilon \sim \mathcal{L}(0, \sigma)$, is a zero-mean Laplace distribution with standard deviation of $\sqrt{2}\sigma$ to be inferred. The prior distribution of σ is specified as a half-normal with scale parameter of unity. The Laplace error model specifies an exponentially decaying distribution (known in Bayesian statistics as the likelihood) of the observed data about our linear model. Here, a Laplace likelihood, rather than the typically selected normal likelihood, is chosen to ensure the regression is more robust to outliers,³⁵ reducing the chance for false positives. However, we note that the posterior distribution of regression coefficients A appears to be insensitive to this choice, indicating that our dataset does not contain highly influential outliers.

The vector A will provide information on linear correlations between inputs and outputs, while b is an offset parameter that we will not need in our assessment. To report the statistical significance of the coefficients, we plot error bars computed from the 2.5% – 97.5% credible interval³⁶ and the smaller of two one-sided credible intervals for each parameter is computed and reported on plots. For example, if a significance level of $S=0$ is reported, that means that the marginal posterior distribution of that parameter lies completely on one side of the null value of 0 for that coefficient, indicating high significance. A one-sided significance level of $S=0.2$ indicates that 20% of the posterior density is on one side of the null result and 80% on the other. Two examples are shown in Fig. 11 to aid in understanding. We note that the one-sided interval definition is chosen to relate to standard frequentist p-values and have a similar interpretation. However, by utilizing a Bayesian approach, we gain direct access to the posterior distribution on the parameters $P(A|X, Y)$. By investigating the structure of this posterior, one may ascertain information about important collinearities in the input data that may mean an effect is difficult to attribute to one or the other input.

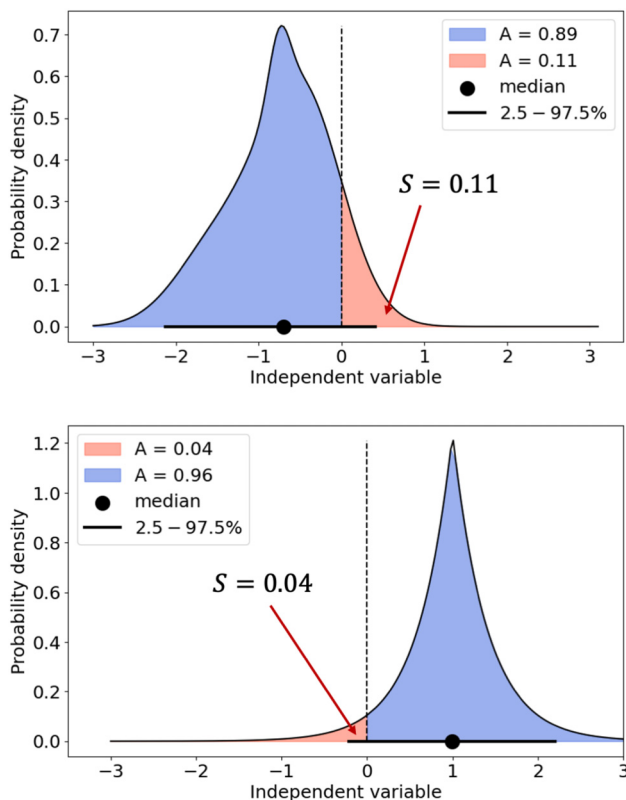


FIG. 11. Each panel demonstrates the computation of the one-sided significance level S for the shown distribution from the smaller of the area A under the curve to the left or right of zero indicated by red shading. The median and 2.5% – 97.5% credible intervals used for plotting in Sec. IV are also shown by the black circle and line respectively. Note that even with a one-sided significance of $S < 0.05$, the credible interval represented by the black “error bar” can contain zero (i.e., the null result).

For our analysis, we apply min–max scaling to all of our data. In other words, all inputs and outputs are normalized to lie in the interval $[0, 1]$. This scaling allows direct comparison of the magnitude of regression coefficients for each input and a given output. Comparison between different outputs is not meaningful. We establish wide priors following a normal distribution $P(\cdot) = \mathcal{N}(0, 2)$ with a standard deviation of $\Sigma = 2$ in our scaled units and a mean $\mu = 0$ centered on the null result to avoid biasing our inference through informative priors. In all cases, we find the posterior distribution is well contained within the prior, indicating that our choice of prior is not influencing our results.

IV. RESULTS AND DISCUSSION

We organize our discussion of the regression results based on the parameter groupings given in Sec. II A. As our approach is intended to contribute to the development of hypotheses for further testing in both simulation and experiment, we do not have a hard threshold for regression coefficients to be significant. Rather, we report the significance level and leave it to future work to determine whether proposed mechanisms do in fact play a significant role in determining morphology. Regardless, we emphasize that the presence or lack of

“significance” does not necessarily indicate that a particular variable is important or unimportant in contributing to the morphology. Indeed, the following are possibilities that may occur when a null result appears likely:

1. Imperfect characterization of experimental data can obscure correlations arising from physical effects.
2. There is no linear correlation and no physical mechanism is expected.
3. There is no linear correlation but non-linear correlations are present with expected physical mechanisms.
4. Input correlations lead to an inability to discern effect.

Unfortunately, quantified uncertainties for much of our data are not readily available. As a result, we cannot directly assess point 1 above or include the impact of measurement uncertainties on the statistical significance of our results at this time. However, our analysis can still be taken to point out interesting areas for further study with this caveat in mind. Investigation of points 2 and 3 above will require significant additional work beyond the scope of this article and is left to future work. For example, by combining inputs using physical insight, e.g., via combinations of dimensionless parameters characterizing various energetics and loss mechanisms of this system,^{23,37} one might capture non-linear correlations without drastically increasing dimensionality. However, even with a simple linear regression, we find evidence supporting existing hypotheses, helping to validate the approach. We also find quantitative evidence consistent with previously unexplored mechanisms impacting morphology, laying the foundation for further exploration.

Point 4 in the list immediately above can arise from the way in which our experiments are designed. Statistical experiment designs with the goal of *exploring* the full parameter space generally seek to sample parameters approximately uniformly throughout the design space region of interest. However, due to the high dimensionality of the experiment input parameter space and the costly nature of MagLIF experiments, such exploratory studies are prohibitively expensive. Instead, our experiments follow a strategy of seeking to understand mechanisms associated with the highest impact single parameter changes to the system in order to improve performance. As a result, our dataset has built-in correlations in the input space. For example, due to susceptibility to growth and feedthrough of MRTI resulting in poor reproducibility, targets with $AR = 9$ have almost exclusively been fielded in a mass-matched dielectric coated configuration, while those with $AR < 9$ are typically not coated. This manifests as a large negative linear correlation between initial outer liner radius and coating (see Fig. 12) since many of our $AR = 9$ liners have a fixed inner radius of 2.325 mm. Typically, positive correlations in the input space allow for increasing the value of the regression coefficient for one input while decreasing it for the other correlated quantity. Similarly, when two inputs are negatively correlated, one may simultaneously increase both coefficients while not impacting the outcome (when all other independent variables are held fixed). Understanding these input correlations and their implications for our regression analysis may point to experiments that could be done to break these correlations to help provide more constraining data. Simulations may also help to physically inform our interpretation. Because the number of pairwise correlations to explore is proportional to the product of the square of the input design space dimension (here there are 18 input variables) with the

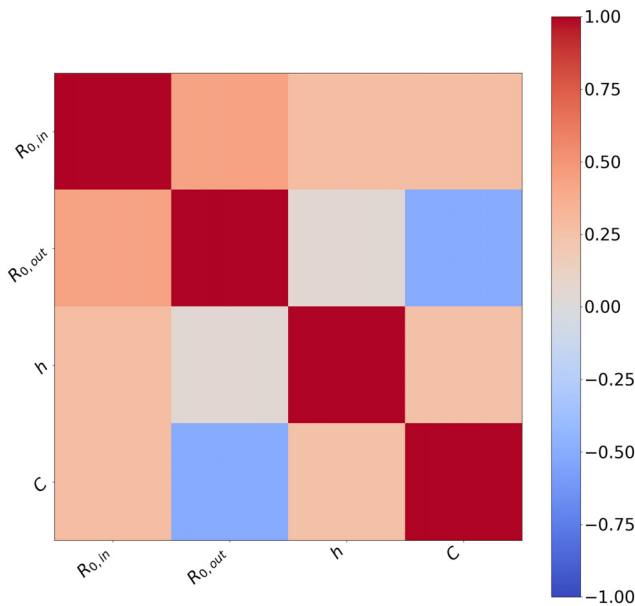


FIG. 12. Correlations between liner parameters can indicate the possibility that certain regions of parameter space are experimentally unexplored. Here, for example, it is shown that the application of dielectric coating is negatively correlated with outer liner radius. This results from the fact that experiments where a dielectric coating has been applied are almost exclusively $AR = 9$ targets.

output space dimension (in this case 3 scalar parameters characterizing morphology), we will not cover these correlations in significant detail. However, when such correlations are expected to at least partly explain our results, we shall mention them briefly in the text. In addition, we include in the supplement figures containing the full linear correlation coefficient matrix of our input values and corner plots showing pairwise marginal posterior distributions of the regression coefficients. This will allow the interested reader to further explore implications of correlations in detail beyond what can be covered here.

Before proceeding, we note that out of the more than 50 regression coefficients explored, a few are likely to falsely show or lack statistical significance (false positive/negative, respectively) due to a combination of the factors 1–4 listed at the beginning of this section. While we attempt to explain the data as observed, we expect that additional detailed investigation will likely show that a small number of our proposed physical mechanisms for the observed correlations are not plausible. In such cases, it may then be useful to revisit the analysis presented here to understand which parts of the experimental parameter space are underexplored and likely to add additional constraining information.

A. Liner parameters

1. E_s (kink-like structure)

Figure 13 shows the inferred correlation strength between liner parameters and morphology measured by E_s , $\overline{CR}$, and σCR in panels (a)–(c), respectively. As seen in Fig. 13(a), the initial inner and outer liner radii show significant positive and negative correlation with E_s , respectively. The observed correlation is consistent with the fact that increasing $R_{0,in}$ or decreasing $R_{0,out}$ will increase the liner aspect ratio defined by $AR = R_{0,out}/(R_{0,out} - R_{0,in})$. This reduces the liner mass, allowing it to achieve greater acceleration, which might contribute to greater growth of MRTI. In addition, the reduced liner thickness will make the liner more susceptible to feedthrough of MRTI since the feedthrough factor is proportional to $\exp(-\Delta/\lambda)$ for liner thickness $\Delta = (R_{0,out} - R_{0,in})$ and MRTI wavelength λ as discussed in Refs. 25 and 38. We note that the correlations between liner inner and outer radii and E_s are consistent with the results shown in Ref. 25 for changing liner aspect ratio.

Liner height may be weakly correlated with E_s , but a null result cannot be ruled out. Applying a mass-matched dielectric coating as described in Ref. 25 shows a significant improvement in the stability metric on average. This result is consistent with the expectation that the coating mitigates growth of ETI that can seed MRTI, as was demonstrated by the analysis presented in Ref. 25 on a small dataset. The

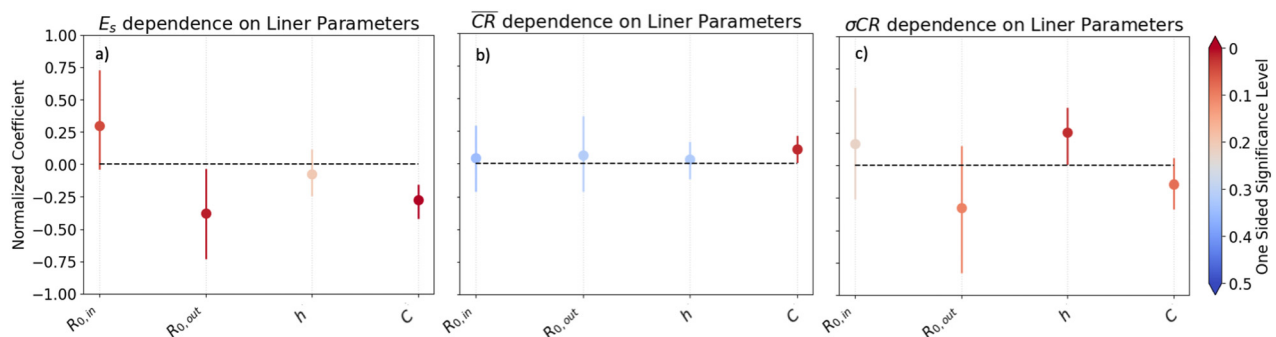


FIG. 13. Correlation coefficients between morphology metrics and liner parameters. (a) Inner and outer liner radii ($R_{0,in}$ and $R_{0,out}$) show strong positive and negative correlation with kink-like structure measured through E_s . This observation is consistent with greater feedthrough of MRTI at higher liner aspect ratios. The negative correlation with dielectric coating C is consistent with reduced growth and feedthrough of MRTI as indicated in Ref. 25. (b) Surprisingly, average convergence $\overline{CR}$ ratio does not appear correlated with either $R_{0,in}$ or $R_{0,out}$, while a dielectric coating is positively correlated with convergence. This may indicate that 3D effects are playing a dominant role. (c) Sausage-like structure characterized by σCR shares positive correlation with $R_{0,in}$ and negative correlation with $R_{0,out}$. Correlation with coating is negative, again consistent with worse MRTI feedthrough at higher liner aspect ratio and mitigation of MRTI by dielectric coatings. The cause for positive correlation with liner height is unexpected and requires further study.

fact that this trend is present across a much larger dataset demonstrates that the effect is statistically robust. In addition, the consistency of our method with the analysis in Ref. 25 for both aspect ratio and coating effects provides additional confidence in our regression method, which is in line with existing experiment analysis.

2. $\overline{CR}$ (average compression)

The apparent null correlation between $\overline{CR}$ and liner radii parameters shown in Fig. 13(b) is not in agreement with intuition. For example, one-dimensional simulations generally show that CR will increase with aspect ratio via increasing $R_{0,in}$ or decreasing $R_{0,out}$ while holding the other fixed. This suggests one of several possibilities. First, it could be that the effect is not measurable within instrument resolution. We consider this possibility to be unlikely since the horizontal imager resolution is typically at or below the strand radius scale, and the average impact of resolution on CR can be quantified reasonably accurately with statistical significance (see Sec. IV D 2). Second, it could be that our data do not sufficiently explore the design space to observe this effect. Again, this seems somewhat unlikely since inner and outer radii vary across a range of hundreds of micrometers and several aspect ratios and are observed to impact the other two morphology parameters with reasonable statistical significance. Finally, it is possible that the convergence ratio is dominated by three-dimensional effects not captured in the one-dimensional simulations. These could include, for example, end losses or growth and feedthrough of MRTI that would affect compression. Interestingly, coatings show a statistically significant positive correlation with $\overline{CR}$, indicating that compression is improved by the presence of coatings. This result is consistent with three-dimensional effects dominating over liner radii effects due to increased feedthrough of MRTI at higher aspect ratios and their mitigation via coatings. It will be interesting to investigate the relative importance of liner inner and outer radii vs growth and feedthrough of MRTI in future work to better understand this surprising null result.

3. σCR (sausage-like structure)

The final case to be considered for liner parameters is the correlation with σCR shown in Fig. 13(c). As discussed in Sec. II B, we intuitively expect σCR to be primarily sensitive to axial variations in compression. As such, we might expect that higher aspect ratios would result in larger values of σCR due to greater potential for feedthrough of MRTI. While the correlation with $R_{0,in}$ is consistent with this intuition, we note that there is a relatively low statistical significance, so the result could be consistent with the null hypothesis that $R_{0,in}$ does not correlate. However, the outer radius shows a more statistically significant regression coefficient, in line with our expectations.

Liner height h shows a significant effect, with taller liners showing greater variation of the convergence ratio. This could be related to a larger initial inductance, the potential for increased participation of power-feed plasma^{39–41} that may impact current delivery as well as initial seeding of MRTI,^{42–45} or other mechanisms that may modify instability growth. It is interesting to note that of the four liner parameters, liner height shows the most statistically significant correlation. The stark contrast with the correlation between h and E_s , which shows a weak or null correlation, indicates that further investigation is

warranted to understand the correlation between liner height and morphology.

Finally, we observe that coatings show a negative correlation, indicating improved uniformity of the implosion by mitigating ETI and subsequent growth of MRTI. The similarity in the signs of correlation coefficients between liner parameters and E_s and σCR is also interesting to note. Despite this similarity, there are significant differences. For example, coating and outer liner radius have the most statistically significant and strongest correlation with E_s , while liner height shows the most statistically significant correlation with σCR among the liner parameters and competes with the other three parameters in correlation strength. Quantitatively understanding the expected relative size of different effects in comparison to our observations will be an interesting topic for further study.

B. Integration parameters

1. E_s (kink-like structure)

Figure 14 shows the correlation strength between integration parameters and morphology metrics. We begin by considering results for the stability metric E_s in Fig. 14(a). There we see that the initial gas fill density and preheat energy deposited appear to be uncorrelated with E_s .

Magnetic field strength shows a reasonably strong negative correlation with E_s . The fact that increased magnetic field strength appears to improve MagLIF stagnation morphology in integrated experiments has not been previously discussed in the literature. However, this is consistent with the observed behavior of other z-pinch plasmas^{46–48} showing that axial magnetic fields act to stabilize current-carrying plasma columns. The result is also qualitatively consistent with theoretical analyses in Refs. 49 and 50 as well as radiography of MagLIF liner implosions without laser preheat.¹³

The final independent variable in this parameter grouping is the peak current. The data show that increasing peak current results in greater E_s on average. This is consistent with expectations that higher peak current will correspond to more aggressive liner acceleration and therefore more significant growth of MRTI when holding all other parameters fixed.²⁷

2. $\overline{CR}$ (average compression)

Correlation between $\overline{CR}$ and integration parameters appears in Fig. 14(b). Surprisingly, fill density and preheat energy do not show correlation with $\overline{CR}$. These null results are counter to expectations from one-dimensional simulations, which show, for example, that increasing fill density across the range that has been fielded experimentally should reduce the convergence ratio by roughly 10%–20% when holding other parameters fixed. For a typical stagnation radius of 50 μm , this corresponds to a change in stagnation radius of $\sim 5 - 10 \mu\text{m}$, which should be resolvable. Similarly, increasing preheat energy should reduce convergence achieved according to 1D simulations.

On the other hand, the magnetic field shows a negative correlation with $\overline{CR}$. The fact that B_0 negatively correlates with $\overline{CR}$ is somewhat unexpected. While it is true that increasing field strength will increase magnetic pressure, we can estimate the plasma β parameter, which in convenient units is given by

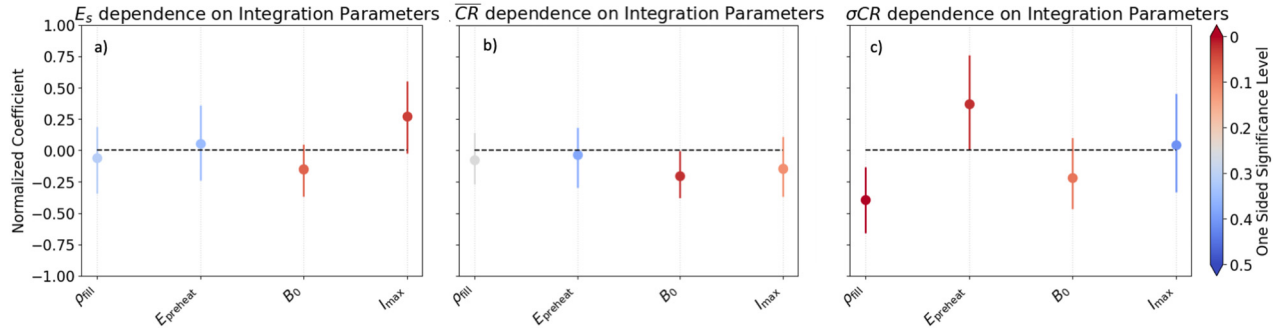


FIG. 14. Correlation coefficients between morphology metrics and integration parameters. (a) Kink-like structure measured by E_s shows negative correlation with the initial applied magnetic field. This constitutes the first observation that axial magnetic field B_0 has a stabilizing effect in integrated MagLIF experiments. The positive correlation with current is expected since increasing peak current will result in more aggressive MRT growth and feedthrough. (b) Average convergence ratio $\overline{CR}$ interestingly shows a null result with preheat, which is expected to negatively correlate according to 1D calculations. The observed negative correlation with initial applied magnetic field is also unexpected, while the negative correlation with peak current is opposite to what one would expect from 1D simulations. This may indicate that three-dimensional effects dominate overall compression. (c) Sausage-like structure σCR shows negative correlation with fill density ρ_{fill} and positive correlation with E_{preheat} . While the reason for this is not clear, it may relate to details of the laser–wall interaction. Interestingly, evidence for magnetic field stabilization of sausage-like structure is observed via the negative correlation similar to the E_s – B_0 correlation. Equally interesting is the fact that peak current does not appear to correlate with this metric.

$$\beta = \frac{4.81 \times 10^8 (\rho [\text{g/cc}]) (T [\text{keV}])}{A(B [\text{kT}])^2}. \quad (10)$$

This quantity measures the ratio between thermal and magnetic pressure near stagnation. In Eq. (10), ρ is the mass density of deuterium fuel at stagnation (in g/cc), T is the ion temperature at stagnation (in keV), A is the average atomic number for the fusion fuel ($A = 2$ for a pure DD plasma), and B is the magnetic field strength at stagnation (in T). For typical values at stagnation are $\rho \approx 1$ g/cc, $T \approx 3$ keV, and $B \approx 10$ kT, which gives $\beta \approx 7.22$ for typical stagnation conditions, indicating that the thermal pressure is $\sim 7\times$ larger. One therefore might not expect magnetic pressure to play a significant role. However, as will be discussed in Sec. IV B 3, B_0 is also observed to be negatively correlated with σCR . It may be that by reducing variation in the convergence ratio, a larger magnetic field strength will result in a reduced average value of the convergence ratio. Furthermore, exploration is warranted to understand the dominant mechanisms determining the observed correlation between B_0 and $\overline{CR}$.

The apparent negative correlation between $\overline{CR}$ and I_{max} is also surprising. In general, one would expect that by increasing the magnetic pressure driving the implosion, a higher convergence would be achieved. However, if 3D effects become more dominant at higher current, compression may actually not strongly increase with current. We also note that the statistical significance is relatively moderate, so a null correlation cannot be ruled out.

Here, we pause to note another possible explanation for the surprising correlation structure between integration parameters and $\overline{CR}$ (specifically the counterintuitive correlation coefficients for E_{preheat} and I_{max}). To this end, we note that improvements to current delivery, applied magnetic field, preheat energy, and fill densities have generally developed in tandem. It is possible that the resulting input correlations mean that the design space is underexplored. This could lead to significant correlations in the posterior distribution. To better demonstrate this possibility, consider the pairwise posterior distribution of the regression coefficients for E_{preheat} and I_{max} with $\overline{CR}$ shown in Fig. 15(a). This plot shows that it is still possible to reasonably describe

the observed data while allowing the regression coefficient $A_{E_{\text{preheat}}}^{\overline{CR}}$ between E_{preheat} and $\overline{CR}$ to become more negative as long as the coefficient $A_{I_{\text{max}}}^{\overline{CR}}$ characterizing correlation between I_{max} and $\overline{CR}$ becomes more positive. Figure 15(b) shows that there are regions of input space at high preheat energy and low current as well as high current and low preheat energy that are unexplored in our dataset. If a sufficient number of experiments were collected in one or both of these regions, it is possible that the posterior would be better constrained along the direction indicated by the red arrow in Fig. 15(a). The green rectangles in Fig. 15(a) show several notional possibilities for how additional data might further constrain the posterior. More detailed investigation into these surprising results for the correlation between $\overline{CR}$ with integration parameters should be pursued.

3. σCR (sausage-like structure)

The third morphology parameter σCR shows reasonably significant correlations with all of the integration parameters except peak current, as seen in Fig. 14(c). The strong negative correlation with ρ_{fill} is unexpected, particularly in light of the null result for ρ_{fill} observed in Fig. 14(b).

Preheat energy E_{preheat} is seen to be positively correlated with σCR . Mechanisms that might explain this include interaction of the laser with the liner inner wall as well as the potential for increased initial pressure to modify acceleration history.

For σCR , increases to the initial applied axial magnetic field are observed to reduce variation in the stagnation convergence ratio. This is consistent with the interpretation provided in discussing the correlation between B_0 and E_s in Sec. IV B 1, indicating that the axial magnetic field has a stabilizing effect on Z-pinch implosions. Interestingly, Refs. 49 and 50 show that increased axial magnetic field strength helps to reduce the growth of sausage-like instabilities during the deceleration phase. Reduction in growth of the sausage instability should reduce σCR in line with our observation of a negative correlation between B_0 and σCR . Although this matches expectations, we emphasize again that our analysis represents the first experimental evidence

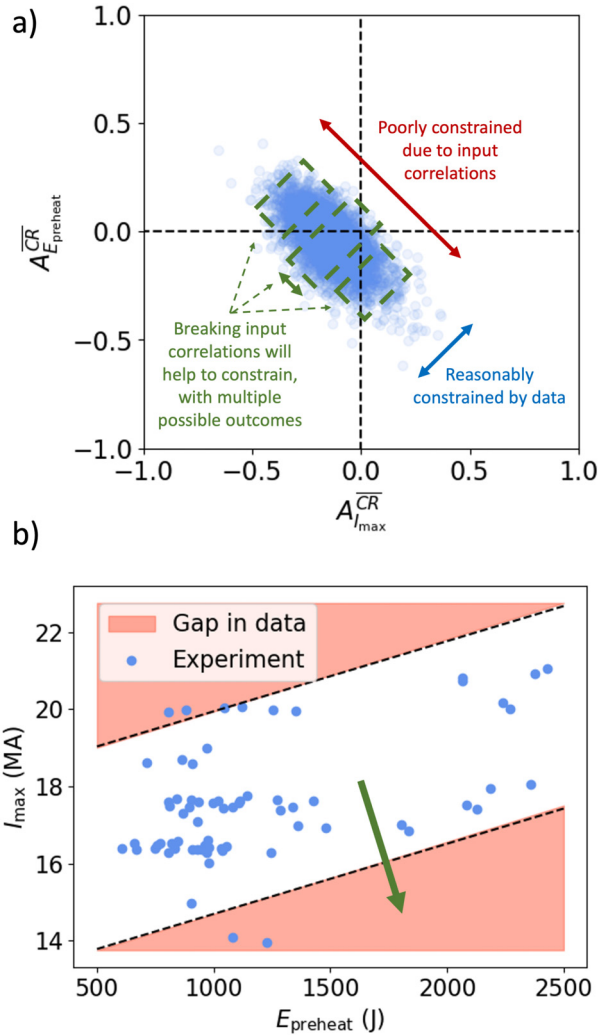


FIG. 15. (a) Correlations between the inputs E_{preheat} and I_{max} appear to be important when determining correlation with CR . In particular, the data allow for some trade-offs between the effect of current and preheat energy on CR . The result is that the posterior is poorly constrained along the axis indicated by the red arrow. By performing experiments to break the input correlation, the posterior should be better constrained along this direction. We show several possible notional outcomes in the green dashed rectangles. (b) Our dataset (with jitter added to allow individual experimental points to be seen) contains significant input correlations. The strongest correlation with I_{max} in the input space is E_{preheat} . To break these correlations, one could perform experiments with parameter values near the head of the green arrow, which increase preheat and reduce current (relative to average). Alternately, increasing current while reducing preheat would also suffice to reduce correlations. Were such experiments performed, the pairwise posterior distribution for $A_{E_{\text{max}}}^{CR}$ and $A_{E_{\text{preheat}}}^{CR}$ shown in panel a might be concentrated in one of the notional green dashed rectangles.

in support of the stabilizing effect of magnetic fields in integrated MagLIF implosions.

Finally, we note that no correlation is found between peak current and σCR . This is strongly counter to intuition, as we would expect that increasing peak current would lead to greater feedthrough of MRTI. It

seems possible that input correlations may be able to explain this surprising null result, but more work is needed to determine the underlying reason.

C. Preheat and magnetic drive protocol parameters

1. E_s (kink-like structure)

The linear correlation between protocol parameters and E_s is shown in Fig. 16(a). The beam profile conditioning does not show an effect on E_s . The LEW thickness shows a negative correlation with E_s , with moderate statistical significance. While a null result cannot be ruled out, it may be worth investigating whether the negative correlation could result from different axial gradients in energy deposition and/or modification of the interaction of the laser preheat blast wave with the inner liner surface. It is also worth noting that E_{preheat} and Δ_{LEW} exhibit significant input correlation that may play a role in the observed results.

The remaining protocol parameters relate to large-scale azimuthal drive asymmetries. As a preface to our discussion, we note that correlation coefficients associated with magnetic drive protocol parameters $C_{m,a}$ and $c_{m,a}(\phi)$ $m = 1, 2, 3$ tend to exhibit larger error bars for all morphology metrics than other coefficients in our study, as seen in Fig. 16. This should not be surprising, as our experiments have not generally been targeted to study the impact of return can asymmetry on morphology. Only 13 of the 57 experiments had asymmetric return cans, and 6 different configurations have been fielded. Despite the large error bars, we will see that a number of coefficients still show relatively high statistical significance. As a result, although the data show strong evidence for an overall trend, the exact size of the effect cannot be accurately determined. Given the low number of repeat experiments and wide variety in return can structures, it is surprising that we find such evidence with any statistical significance at all. This highlights the power of our method to seek new hypotheses from existing data to inform future experiments.

For the case of E_s , the only magnetic drive protocol parameter that correlates with reasonably strong statistical significance is the line-of-sight-dependent mode 1 azimuthal drive factor $c_{1,a}(\phi)$. It is not presently clear how drive asymmetry as measured by $C_{m,a}$ and $c_{m,a}(\phi)$ should be expected to impact E_s . However, considering the quasi-orthogonal views of z3731 in Fig. 3, it seems plausible that low-mode drive asymmetry can give rise to axially coherent “pancaking” of the stagnation column. The left of these two images had $c_{1,a}(\phi_1) = -0.81$, while the second view had $c_{1,a}(\phi_2) = -0.18$. In Sec. IV C 2, we will see evidence of correlations between $c_{1,a}(\phi)$ and CR that indicate the possibility that certain combinations of drive asymmetry may set up this “pancaked” structure. We note that when such structure persists over large portions of the stagnation column, it is reasonable to presume that growth rates of kink-like structure along the major and minor axes will differ due to the breaking of cylindrical symmetry. The authors are not aware of any current simulation or theory-based analysis that directly supports this claim in the plasma regime relevant for MagLIF. However, linear analysis of the kink mode for plasma columns with elliptical cross sections has shown an asymmetry in dispersion relations along major and minor ellipse axes for sufficiently large ellipticity.⁵¹ It seems plausible that the development of large-scale coherent “pancaking” of our plasma column may then result in asymmetric development of kink structure. This apparent

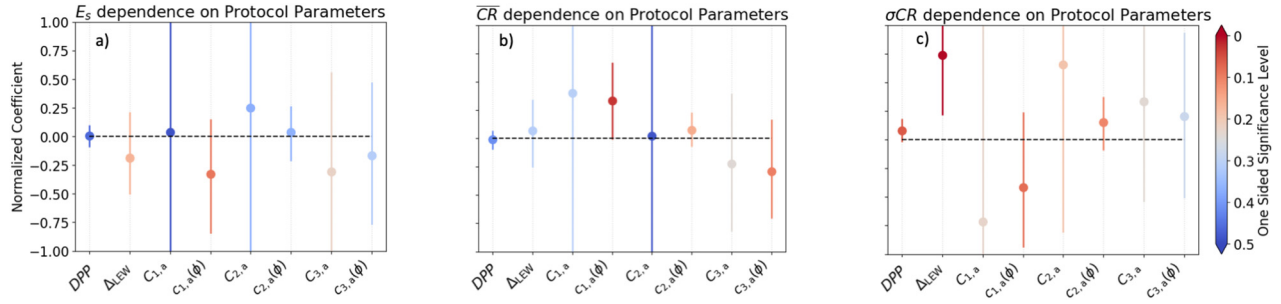


FIG. 16. Correlation coefficients laser preheat and magnetic drive protocol parameters and (a) kink-like structure measured by E_s , (b) average convergence ratio $\overline{CR}$, and (c) sausage-like structure measured by σCR . Interestingly, all three image metrics show correlations among various combinations of the drive asymmetry amplitudes and view factors. This is the first time that drive asymmetry has been demonstrated to impact stagnation morphology in integrated MagLIF experiments. The variation in convergence ratio σCR also shows a strong positive correlation with LEW thickness Δ_{LEW} as seen in panel (c). This may be a result of the large-scale axial variation in energy deposition depending on LEW thickness as indicated in Ref. 18.

feature in our data encourages further investigation into the underlying physical mechanisms relating drive asymmetry to E_s .

2. $\overline{CR}$ (average compression)

Figure 16(b) demonstrates that the average convergence ratio does not appear to be significantly influenced by either of the two pre-heat protocol parameters of beam conditioning or LEW thickness. This may indicate that any three-dimensional effects associated with these parameters do not strongly affect overall compression.

The drive asymmetry mode amplitudes $C_{m,a}$ do not show statistically significant correlations with $\overline{CR}$ but have large uncertainty. Additional experimental data targeting this effect may help to provide constraint. In regard to view factor amplitudes, $c_{1a}(\phi)$ shows a statistically significant positive correlation with $\overline{CR}$, while $c_{m,a}(\phi)$ for $m = 2, 3$ shows moderately significant correlations, indicating the possibility for a complex interplay of different modes in compressing the target. Detailed simulation and theoretical analyses are warranted in the future to understand the impact of drive asymmetry on compression.

3. σCR (sausage-like structure)

The dependence of σCR on protocol parameters is shown in Fig. 16(c). Although beam conditioning appears to show a statistically significant effect, the magnitude is rather low. On the other hand, Δ_{LEW} shows a statistically significant effect of high magnitude. Referring back to Fig. 2(d) a thicker LEW has been observed to reduce laser propagation depth. Indeed, Refs. 18 and 20 experimentally infer the energy deposition vs depth. It seems plausible that large-scale variation in laser energy deposition could lead to large-scale variation in convergence ratio that would result in an increase to σCR . Furthermore, Fig. 12 of Ref. 18 shows that LEW thickness Δ_{LEW} has a much stronger impact on the axial distribution of energy deposited than the beam conditioning. The larger correlation coefficient for Δ_{LEW} as compared to DPP and the increased gradient in energy deposition with increasing Δ_{LEW} are both consistent with the observed correlations in our dataset. Regardless of whether this intuitive explanation is responsible, this is the first time that direct experimental evidence has been provided for the impact of LEW thickness on observed MagLIF stagnation morphology. Additionally, we emphasize

that this is just one of many possible effects, such as end losses and growth of MRTI, likely to contribute, so that it is not clear *a priori* that an effect from the LEW should be observable in our data.

The view factor amplitudes $c_{m,a}(\phi)$ for $m = 1, 2$ show moderately significant correlations with σCR . It is not clear that such correlations should be expected. However, these correlations, paired with similar observations for E_s and $\overline{CR}$, encourage more detailed study to explore how of azimuthal drive asymmetry interacts with growth of MRTI, compression efficiency, and development of stagnation structure.

D. Diagnostic parameters

1. E_s (kink-like structure)

The linear correlation between diagnostic parameters and E_s is shown in Fig. 17(a). In this case, no significant correlation is observed with respect to horizontal imager resolution. In retrospect, this should be anticipated. In particular, imager resolution in the horizontal dimension varies between roughly $15 - 70 \mu\text{m}$, which is comparable to or below the strand radius. Since the metric E_s is computed from a mean of contours that attempts to characterize the left and right boundaries of the strand at a particular height as independently of brightness variation across the strand as possible, it is relatively insensitive to radial structure in the strand. Any blurring in the horizontal dimension is likely to result in fairly symmetric widening of the strand so that the average position will not be significantly altered unless simultaneously averaging over height due to vertical resolution causes significant modification. The only impact should therefore be expected if vertical resolution is of similar or larger length scale as the axial length scale associated with kink-like structure. Recalling that the feed-through factor for MRTI scales as $\exp(-\Delta/\lambda)$, where Δ is the liner thickness and λ is the axial wavelength of the particular MRTI mode being considered.³⁸ This results in natural filtering of short wavelength modes, with experimentally observed typical length scales of kink-like structure being on the order of $\lambda \sim 0.5 - 3 \text{ mm}$. The largest vertical imager resolution is $V_{\text{res,max}} = 0.120 \text{ mm} \approx \lambda_{\text{min}}/4$. Since the largest value of vertical imager resolution approaches the lower end of typical kink-structure length scale, a small effect from V_{res} is not unreasonable. We also note that the pincushion effect mentioned in Sec. II A 4 is generally enhanced for imager configurations with larger values of

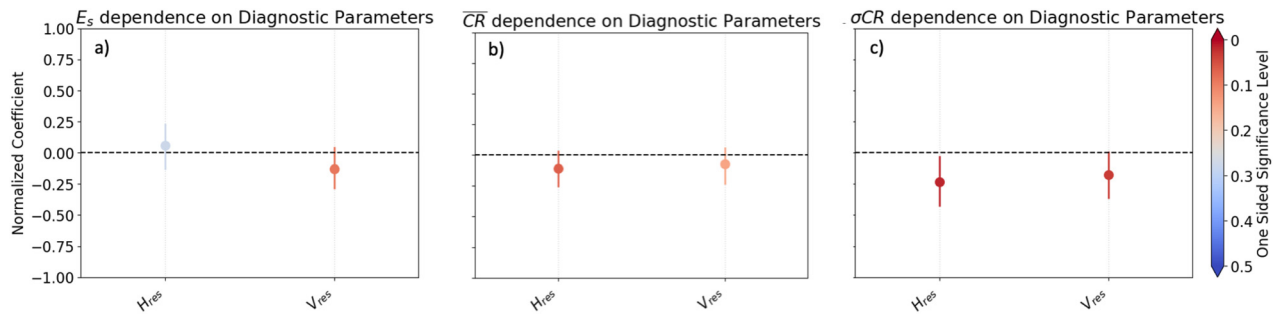


FIG. 17. Correlation coefficients between morphology metrics and diagnostic parameters. (a) Only the vertical imager resolution V_{res} appears to correlate with E_s . While the reason for such a correlation is not immediately obvious, a plausible explanation is provided in the main text. (b) Horizontal imager resolution H_{res} shows a reasonably significant correlation with $\overline{CR}$ as expected, while a possible negative correlation with V_{res} is plausible as described in the main text, although the similar magnitude of the two correlation coefficients is somewhat surprising. (c) Both vertical and horizontal imager resolution appear to correlate negatively with σCR . The result for H_{res} is expected and can be intuitively explained by considering the limit of large H_{res} where the strand width will be completely determined by imager resolution. Reasoning for the V_{res} correlation is similar to that for the correlation between $\overline{CR}$ and V_{res} as described in the main text. In all cases, the correlation of diagnostic resolution with each of the image metrics is reasonably aligned with intuition based on whether horizontal and vertical blurring at the scale of diagnostic resolution is similar to feature scales in the image metrics.

horizontal and vertical resolution. The pincushion effect may result in local reduction of amplitude in regions where the mean strand position has curvature opposite to the pincushion distortion in the PSF.

2. $\overline{CR}$ (average compression)

As opposed to the case for E_s , one may *a priori* expect a significant linear correlation between imager resolution and $\overline{CR}$. Qualitatively, imager resolution in the horizontal dimension varies from a sub-strand radius scale of $15\ \mu\text{m}$ up to about $70\ \mu\text{m}$, which is similar in scale to the expected strand radius of about $50\ \mu\text{m}$ and may cause significant apparent widening of the strand. Figure 17(b) shows that this is indeed the case, with the negative sign for the correlation between $\overline{CR}$ and H_{res} indicating that worse resolution, i.e., larger resolution length scale, will blur the image in the horizontal dimension, resulting in a larger strand width, and lowering the apparent convergence ratio. The presence of a negative correlation with the vertical resolution is perhaps less intuitive, but similar to the case for E_s , if the vertical resolution is of a similar scale to the length scale corresponding to variations in CR across the axial height of the strand, then one would expect additional reduction of $\overline{CR}$. Variations in $CR(z)$ occur on a variety of length scales from $\sim 500\ \mu\text{m} - 4\ \text{mm}$. The largest value of V_{res} for imagers used in our dataset approach the smallest length scale for variation in CR . This may explain the observed negative sign for the regression coefficient, but a null result is also possible. We consider the fact that correlation of $\overline{CR}$ with vertical resolution is of a similar magnitude as the correlation between $\overline{CR}$ and H_{res} to be somewhat surprising but plausible given the above argument. We note that although it has been known that the imager resolution will impact our quantification of image morphology, to date, care has been taken to ensure that only images with similar resolution have been quantitatively compared. By including imager resolution as an independent variable in our regression, we *quantitatively* estimate and remove the impact of imager resolution to allow for reasonably accurate direct comparison between images without the need for complicated deconvolution methods.

3. σCR (sausage-like structure)

The morphology measured by σCR demonstrates correlation with imager resolution for similar reasons as $\overline{CR}$. Considering the limiting case of large H_{res} , the strand width will be completely determined by the resolution, thereby reducing the variation. This argument is consistent with the observed correlation indicating that larger horizontal imager resolution reduces apparent variation in CR . Reasoning for dependence on V_{res} is the same as for $\overline{CR}$. Finally, it is interesting to note that although vertical and horizontal resolutions are positively correlated in our dataset, the correlation is not perfectly linear. We are therefore able to see the impact that both have on estimating $\overline{CR}$ as well as σCR .

V. CONCLUSION AND FUTURE WORK

Self-emission x-ray images show rich structure in MagLIF stagnation columns that has been attributed to three-dimensional effects. When applying a Bayesian linear regression framework, we have discovered a number of correlations between metrics characterizing the stagnation morphology and MagLIF input conditions and diagnostic setup. Below we summarize several key findings of our work.

1. Inner liner radii, $R_{0,\text{in}}$ and outer $R_{0,\text{out}}$, respectively, appear to correlate with both E_s (kink-like structure) and σCR (sausage-like structure), consistent with increased feedthrough of MRTI at higher aspect ratios. Coatings C negatively correlate with both E_s and σCR , consistent with improved stabilization through mitigation of ETI and MRTI [see Figs. 13(a) and 13(c)].
2. Counter to intuition, inner and outer liner radii $R_{0,\text{in}}$ and outer $R_{0,\text{out}}$ do not show strong correlation with $\overline{CR}$ (average compression) expected from increased compression at higher aspect ratios. Coatings C are observed to increase $\overline{CR}$ [see Fig. 13(b)]. This is consistent with the expectation that coatings reduce three-dimensional effects that can degrade compression and confinement.
3. Preheat energy E_{preheat} and fill density ρ_{fill} do not appear to impact convergence ratio $\overline{CR}$, counter to intuition from 1D simulations. At the same time, I_{max} shows a reasonably significant

- negative correlation with $\overline{CR}$ where positive correlation is expected [see Fig. 14(b)]. These observations may indicate that MRTI dominates over any effect from preheat energy and fill density. However, more work is needed to understand the underlying reason for these observations.
- Preheat energy E_{preheat} has significant positive correlation with σCR [see Fig. 14(c)]. Fill density ρ_{fill} on the other hand shows significant negative correlation with σCR . The cause of these results is not yet known. Our analysis provides the first evidence that preheat energy and fill density impact stagnation morphology, encouraging further investigation.
 - Initial magnetic field strength B_0 negatively correlates with E_s and σCR , indicating stability is improved with increasing B_0 [see Figs. 14(a) and 14(c)]. We note that although the initial field strength is $\mathcal{O}(10\text{ T})$, flux compression results in an $\mathcal{O}(1 - 10\text{ kT})$ axial magnetic field strength at stagnation that is expected to depend in part on initial field strength.⁸ This is the first time that evidence for the stabilizing effect of magnetic tension has been observed in integrated MagLIF experiments.
 - Asymmetric return can structure measured through $C_{m,a}$ and $c_{m,a}(\phi)$ ($m=1,2,3$) is expected to result in drive asymmetries.⁵² Correlations with these magnetic drive protocol parameters accounting for view angle relative to drive asymmetry indicate this has a measurable impact on morphology. All three of E_s , $\overline{CR}$, and σCR show a variety of correlations that are reasonably statistically significant although with large credible intervals (see Fig. 16). This is the first time that the impact of drive asymmetry on stagnation morphology has been assessed.
 - The LEW thickness Δ_{LEW} shows significant positive correlation with σCR [see Fig. 16(c)]. This may, for example, be a result of axially varying energy deposition from the preheat laser, which has a steeper gradient for thicker LEWs.
 - By incorporating horizontal and vertical imager resolution, H_{res} and V_{res} , respectively, in our regression, we can quantitatively assess the impact of resolution on our image metrics. This allows for fair comparison of images despite differences in resolution, without the need to apply complex deconvolution methods (see Fig. 17).

We note in particular that correlations of E_s with inner and outer liner radii as well as coating are consistent with the results of Ref. 25 analyzing a smaller dataset. This provides some validation of our approach, while also indicating statistical robustness of the analysis in Ref. 25. Our analysis also indicates some effects that have not been previously demonstrated, but which align with expectations. This shows that the simple regression approach is useful for data exploration to highlight interesting correlations. Some counterintuitive null results, including the lack of impact from liner inner and outer radii on $\overline{CR}$ were found. This encourages further investigation into competing mechanisms that may explain the observations. Additionally, we demonstrated that azimuthal drive asymmetries resulting from current return can structure appear to measurably impact the morphology, a feature which has not been previously analyzed. Theoretical and simulation-based studies of this effect are likely to further inform experiment protocols to ensure good implosion performance. Due to the relatively large design space and comparatively small number of experimental images (64 images collected across 57 experiments), one may reasonably expect a small number statistically significant (as well

as apparent null) results may be false positives (negatives). While caution should be used when interpreting results, particularly those with low statistical significance, the observed correlation coefficients provide an interesting approach to data exploration for driving future investigations, even when error bars are large.

Finally, we emphasize that this approach is widely applicable and may be easily adapted to other metrics or platforms. For example, metrics could be computed from full 3D reconstructions of stagnation emission volumes²⁸ rather than the individual projection images themselves as we collect more data on experiments with multiple views. Additionally, it will be interesting to investigate correlations between experimental inputs and metrics computed from other diagnostics such as filtered x-ray pinhole camera images,⁵³ axially resolved x-ray spectra,⁵⁴ one-dimensional neutron images,^{55–57} and neutron time-of-flight detectors.^{58,59} Such analysis may provide new insights, for example, into the most important mechanisms for the introduction of mix from the liner into the fuel. The method could also be easily adapted to study other platforms such as the Argon gas-puff x-ray source fielded on Z.^{60–62} We anticipate that this statistical approach to data exploration and analysis will continue to lead to a deeper understanding of high energy density physics platforms fielded at a variety of facilities.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for several supporting figures and discussion. The supplement provides evidence to justify choices made in the contour finding algorithm for quantifying morphology, expectations regarding sensitivity to these choices, and discussion of the two images that required by-hand treatment. Also include is a plot of the pairwise correlation between input parameters across our dataset as well as corner plots showing all of the pairwise posterior distributions for regression coefficients. This will allow the interested reader to investigate the implications of correlations in the input space on our analysis in more detail than can be described in the main text.

ACKNOWLEDGMENTS

W. Lewis would like to thank the MagLIF working group for useful discussions. Image credit for tomographic slices in Fig. 2(a) to Philip Noell and James Griego of Sandia National Laboratories.

This article has been authored by an employee of National Technology & Engineering Solutions of Sandia, LLC under Contract No. DE-NA0003525 with the U.S. Department of Energy (DOE). The employee owns all right, title, and interest in and to the article and is solely responsible for its contents. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this article or allow others to do so, for United States Government purpose. The DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan <https://www.energy.gov/downloads/doe-public-access-plan>.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

William E. Lewis: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Project administration (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead).

Adam J. Harvey-Thompson: Conceptualization (equal); Data curation (equal); Formal analysis (supporting); Investigation (equal); Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (equal).

Patrick F. Knapp: Conceptualization (supporting); Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Writing – review & editing (supporting).

Owen M. Mannion: Conceptualization (supporting); Data curation (equal); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Writing – review & editing (equal).

Daniel E. Ruiz: Conceptualization (supporting); Formal analysis (equal); Investigation (equal); Methodology (supporting); Writing – review & editing (equal).

Marc-Andre Schaeuble: Conceptualization (supporting); Data curation (equal); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Software (supporting); Writing – review & editing (equal).

Stephen A. Slutz: Conceptualization (supporting); Formal analysis (supporting); Investigation (supporting); Software (supporting); Writing – review & editing (equal).

Matthew R. Weis: Conceptualization (supporting); Formal analysis (supporting); Investigation (supporting); Methodology (supporting); Writing – review & editing (equal).

Jeffrey Woolstrum: Conceptualization (supporting); Investigation (supporting); Software (supporting); Writing – review & editing (supporting).

David J. Ampleford: Conceptualization (supporting); Data curation (supporting); Investigation (supporting); Project administration (equal); Resources (equal); Writing – review & editing (equal).

Luke Shulenburger: Conceptualization (supporting); Investigation (supporting); Methodology (supporting); Project administration (equal); Writing – review & editing (equal).

David A. Yager-Elorriaga: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (equal); Methodology (equal); Software (lead); Writing – original draft (equal); Writing – review & editing (equal).

Christopher A. Jennings: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Writing – original draft (equal); Writing – review & editing (equal).

Jeffrey R. Fein: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Methodology (equal); Writing – review & editing (equal).

Gabriel A. Shipley: Conceptualization (equal); Data curation (supporting); Formal analysis (equal); Investigation (equal); Methodology (supporting); Software (equal); Writing – original draft (supporting); Writing – review & editing (supporting).

Andrew Porwitzky: Conceptualization (equal); Data curation (equal); Investigation (supporting); Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (supporting).

Thomas J. Awe: Conceptualization (equal); Data curation (equal); Investigation (equal); Writing – original draft (supporting); Writing – review & editing (equal).

Matthew R. Gomez: Conceptualization (equal); Data curation (equal); Formal analysis (supporting); Investigation (equal); Methodology (supporting); Writing – original draft (equal); Writing – review & editing (equal).

Eric C. Harding: Conceptualization (supporting); Data curation (equal); Formal analysis (supporting); Investigation (supporting);

Methodology (supporting); Writing – original draft (supporting); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- ¹S. A. Slutz, M. C. Herrmann, R. A. Vesey, A. B. Sefkow, D. B. Sinars, D. C. Rovang, K. J. Peterson, and M. E. Cuneo, “Pulsed-power-driven cylindrical liner implosions of laser preheated fuel magnetized with an axial field,” *Phys. Plasmas* **17**(5), 056303 (2010).
- ²D. Yager-Elorriaga, M. Gomez, D. Ruiz, S. Slutz, A. Harvey-Thompson, C. Jennings, P. Knapp, P. Schmit, M. Weis, T. Awe, G. Chandler, M. Mangan, C. Myers, J. Fein, B. Galloway, M. Geissel, M. Glinsky, S. Hansen, E. Harding, D. Lampa, W. Lewis, P. Rambo, G. Robertson, M. Savage, G. Shipley, I. Smith, J. Schwarz, D. Ampleford, K. Beckwith, K. Peterson, J. Porter, G. Rochau, and D. Sinars, “An overview of magneto-inertial fusion on the Z machine at Sandia National Laboratories,” *Nucl. Fusion* **62**, 042015 (2022).
- ³D. C. Rovang, D. C. Lampa, M. E. Cuneo, A. C. Owen, J. McKenney, D. W. Johnson, S. Radovich, R. J. Kaye, R. D. McBride, C. S. Alexander, T. J. Awe, S. A. Slutz, A. B. Sefkow, T. A. Haill, P. A. Jones, J. W. Argo, D. G. Dalton, G. K. Robertson, E. M. Waisman, D. B. Sinars, J. Meissner, M. Milhous, D. N. Nguyen, and C. H. Mielke, “Pulsed-coil magnet systems for applying uniform 10–30 T fields to centimeter-scale targets on Sandia’s Z facility,” *Rev. Sci. Instrum.* **85**(12), 124701 (2014).
- ⁴P. K. Rambo, I. C. Smith, J. L. Porter, M. J. Hurst, C. S. Speas, R. G. Adams, A. J. Garcia, E. Dawson, B. D. Thurston, C. Wakefield, J. W. Kellogg, M. J. Slattery, H. C. Ives, R. S. Broyles, J. A. Caird, A. C. Erlandson, J. E. Murray, W. C. Behrendt, N. D. Neilsen, and J. M. Narduzzi, “Z-Beamlet: a multikilojoule, terawatt-class laser system,” *Appl. Opt.* **44**, 2421–2430 (2005).
- ⁵P. Rambo, J. Schwarz, M. Schollmeier, M. Geissel, I. Smith, M. Kimmel, C. Speas, J. Shores, D. Armstrong, J. Bellum, E. Field, D. Kletecka, and J. Porter, “Sandia’s Z-Backlighter Laser Facility,” *Proc. SPIE* **10014**, 100140Z (2016).
- ⁶M. R. Gomez, S. A. Slutz, A. B. Sefkow, D. B. Sinars, K. D. Hahn, S. B. Hansen, E. C. Harding, P. F. Knapp, P. F. Schmit, C. A. Jennings, T. J. Awe, M. Geissel, D. C. Rovang, G. A. Chandler, G. W. Cooper, M. E. Cuneo, A. J. Harvey-Thompson, M. C. Herrmann, M. H. Hess, O. Johns, D. C. Lampa, M. R. Martin, R. D. McBride, K. J. Peterson, J. L. Porter, G. K. Robertson, G. A. Rochau, C. L. Ruiz, M. E. Savage, I. C. Smith, W. A. Stygar, and R. A. Vesey, “Experimental demonstration of fusion-relevant conditions in magnetized liner inertial fusion,” *Phys. Rev. Lett.* **113**, 155003 (2014).
- ⁷P. F. Knapp, M. E. Glinsky, M. A. Schaeuble, C. A. Jennings, M. Evans, J. Gunning, T. J. Awe, G. A. Chandler, M. Geissel, M. R. Gomez, K. D. Hahn, S. B. Hansen, E. C. Harding, A. J. Harvey-Thompson, S. Humane, B. T. Klein, M. Mangan, T. Nagayama, A. J. Porwitzky, D. E. Ruiz, P. F. Schmit, S. A. Slutz, I. C. Smith, M. R. Weis, D. A. Yager-Elorriaga, D. J. Ampleford, K. Beckwith, T. R. Mattsson, K. J. Peterson, and D. B. Sinars, “Estimation of stagnation performance metrics in magnetized liner inertial fusion experiments using Bayesian data assimilation,” *Phys. Plasmas* **29**(5), 052711 (2022).
- ⁸W. E. Lewis, O. M. Mannion, D. E. Ruiz, C. A. Jennings, P. F. Knapp, M. R. Gomez, A. J. Harvey-Thompson, M. R. Weis, S. A. Slutz, D. J. Ampleford, and K. Beckwith, “Data-driven assessment of magnetic charged particle confinement parameter scaling in magnetized liner inertial fusion experiments on Z,” *Phys. Plasmas* **30**, 052701 (2023).
- ⁹T. J. Awe, K. J. Peterson, E. P. Yu, R. D. McBride, D. B. Sinars, M. R. Gomez, C. A. Jennings, M. R. Martin, S. E. Rosenthal, D. G. Schroen, A. B. Sefkow, S. A. Slutz, K. Tomlinson, and R. A. Vesey, “Experimental demonstration of the stabilizing effect of dielectric coatings on magnetically accelerated imploding metallic liners,” *Phys. Rev. Lett.* **116**, 065001 (2016).
- ¹⁰P. F. Knapp, M. R. Martin, D. H. Dolan, K. Cochran, D. Dalton, J.-P. Davis, C. A. Jennings, G. P. Loisel, D. H. Romero, I. C. Smith, E. P. Yu, M. R. Weis, T. R. Mattsson, R. D. McBride, K. Peterson, J. Schwarz, and D. B. Sinars, “Direct

- measurement of the inertial confinement time in a magnetically driven implosion,” *Phys. Plasmas* **24**, 042708 (2017).
- ¹¹E. P. Yu, T. J. Awe, K. R. Cochrane, K. J. Peterson, K. C. Yates, T. M. Hutchinson, M. W. Hatch, B. S. Bauer, K. Tomlinson, and D. B. Sinars, “Three-dimensional feedback processes in current-driven metal,” *Phys. Rev. E* **107**, 065209 (2023).
 - ¹²E. P. Yu, T. J. Awe, K. R. Cochrane, K. J. Peterson, K. C. Yates, T. M. Hutchinson, M. W. Hatch, B. S. Bauer, K. Tomlinson, and D. B. Sinars, “Seeding the electrothermal instability through a three-dimensional, nonlinear perturbation,” *Phys. Rev. Lett.* **130**, 255101 (2023).
 - ¹³T. J. Awe, R. D. McBride, C. A. Jennings, D. C. Lamppa, M. R. Martin, D. C. Rovang, S. A. Slutz, M. E. Cuneo, A. C. Owen, D. B. Sinars, K. Tomlinson, M. R. Gomez, S. B. Hansen, M. C. Herrmann, J. L. McKenney, C. Nakhleh, G. K. Robertson, G. A. Rochau, M. E. Savage, D. G. Schroen, and W. A. Stygar, “Observations of modified three-dimensional instability structure for imploding z-pinch liners that are premagnetized with an axial field,” *Phys. Rev. Lett.* **111**, 235005 (2013).
 - ¹⁴E. C. Harding, G. K. Robertson, G. S. Dunham, M. R. Gomez, J. R. Fein, P. F. Knapp, A. J. Harvey-Thompson, C. S. Spears, D. J. Ampleford, G. A. Rochau, R. Doron, and Y. Maron, “X-ray self-emission imaging with spherically bent Bragg crystals on the Z-machine,” *Rev. Sci. Instrum.* **94**, 083509 (2023).
 - ¹⁵A. J. Harvey-Thompson, A. B. Sefkow, T. N. Nagayama, M. S. Wei, E. M. Campbell, G. Fiksel, P.-Y. Chang, J. R. Davies, D. H. Barnak, V. Y. Glebov, P. Fitzsimmons, J. Fooks, and B. E. Blue, “Diagnosing laser-preheated magnetized plasmas relevant to magnetized liner inertial fusion,” *Phys. Plasmas* **22**(12), 122708 (2015).
 - ¹⁶M. Geissel, A. J. Harvey-Thompson, T. J. Awe, D. E. Bliss, M. E. Glinsky, M. R. Gomez, E. Harding, S. B. Hansen, C. Jennings, M. W. Kimmel, P. Knapp, S. M. Lewis, K. Peterson, M. Schollmeier, J. Schwarz, J. E. Shores, S. A. Slutz, D. B. Sinars, I. C. Smith, C. S. Spears, R. A. Vesey, M. R. Weis, and J. L. Porter, “Minimizing scatter-losses during pre-heat for magneto-inertial fusion targets,” *Phys. Plasmas* **25**(2), 022706 (2018).
 - ¹⁷A. J. Harvey-Thompson, M. R. Weis, E. C. Harding, M. Geissel, D. J. Ampleford, G. A. Chandler, J. R. Fein, M. E. Glinsky, M. R. Gomez, K. D. Hahn, S. B. Hansen, C. A. Jennings, P. F. Knapp, R. R. Paguio, L. Perea, K. J. Peterson, J. L. Porter, P. K. Rambo, G. K. Robertson, G. A. Rochau, D. E. Ruiz, J. Schwarz, J. E. Shores, D. B. Sinars, S. A. Slutz, G. E. Smith, I. C. Smith, C. S. Spears, and K. Whittemore, “Diagnosing and mitigating laser preheat induced mix in MagLIF,” *Phys. Plasmas* **25**(11), 112705 (2018).
 - ¹⁸A. J. Harvey-Thompson, M. Geissel, C. A. Jennings, M. R. Weis, M. R. Gomez, J. R. Fein, D. J. Ampleford, G. A. Chandler, M. E. Glinsky, K. D. Hahn, S. B. Hansen, E. C. Harding, P. F. Knapp, R. R. Paguio, L. Perea, K. J. Peterson, J. L. Porter, P. K. Rambo, G. K. Robertson, G. A. Rochau, C. L. Ruiz, J. Schwarz, J. E. Shores, D. B. Sinars, S. A. Slutz, G. E. Smith, I. C. Smith, C. S. Spears, K. Whittemore, and D. Woodbury, “Constraining preheat energy deposition in MagLIF experiments with multi-frame shadowgraphy,” *Phys. Plasmas* **26**(3), 032707 (2019).
 - ¹⁹S. M. Miller, S. A. Slutz, S. N. Bland, S. R. Klein, P. C. Campbell, J. M. Woolstrum, C. C. Kuranz, M. R. Gomez, N. M. Jordan, and R. D. McBride, “A pulsed-power implementation of “Laser Gate” for increasing laser energy coupling and fusion yield in magnetized liner inertial fusion (MagLIF),” *Rev. Sci. Instrum.* **91**, 063507 (2020).
 - ²⁰B. R. Galloway, S. A. Slutz, M. W. Kimmel, P. K. Rambo, J. Schwarz, M. Geissel, A. J. Harvey-Thompson, M. R. Weis, C. A. Jennings, E. S. Field, D. E. Kleitecka, Q. Looker, A. P. Colombo, A. D. Edens, I. C. Smith, J. E. Shores, C. S. Spears, R. J. Spears, A. P. Spann, J. Sin, S. Gautier, V. Sauget, P. A. Treadwell, G. A. Rochau, and J. L. Porter, “Lasergate: A windowless gas target for enhanced laser preheat in magnetized liner inertial fusion,” *Phys. Plasmas* **28**, 112703 (2021).
 - ²¹A. J. Harvey-Thompson, M. Geissel, J. A. Crabtree, M. R. Weis, M. R. Gomez, J. R. Fein, W. E. Lewis, D. J. Ampleford, T. J. Awe, G. A. Chandler, B. R. Galloway, S. B. Hansen, J. Hanson, E. C. Harding, C. A. Jennings, M. Kimmel, P. F. Knapp, M. A. Mangan, A. Maurer, R. R. Paguio, L. Perea, K. J. Peterson, J. L. Porter, P. K. Rambo, G. K. Robertson, G. A. Rochau, D. E. Ruiz, J. E. Shores, S. A. Slutz, G. E. Smith, I. C. Smith, C. S. Spears, D. A. Yager-Elorriaga, and A. York, “Demonstration of improved laser preheat with a cryogenically cooled magnetized liner inertial fusion platform,” *Rev. Sci. Instrum.* **94**, 053506 (2023).
 - ²²M. R. Weis, A. J. Harvey-Thompson, and D. E. Ruiz, “Scaling laser preheat for MagLIF with the Z-Beamlet laser,” *Phys. Plasmas* **28**(1), 012705 (2021).
 - ²³P. F. Schmit and D. E. Ruiz, “A conservative approach to scaling magneto-inertial fusion concepts to larger pulsed-power drivers,” *Phys. Plasmas* **27**(6), 062707 (2020).
 - ²⁴M. E. Glinsky, T. W. Moore, W. E. Lewis, M. R. Weis, C. A. Jennings, D. J. Ampleford, P. F. Knapp, E. C. Harding, M. R. Gomez, and A. J. Harvey-Thompson, “Quantification of MagLIF morphology using the Mallat scattering transformation,” *Phys. Plasmas* **27**(11), 112703 (2020).
 - ²⁵D. J. Ampleford, D. A. Yager-Elorriaga, C. A. Jennings, E. C. Harding, M. R. Gomez, A. J. Harvey-Thompson, T. Awe, G. A. Chandler, G. S. Dunham, M. Geissel, K. D. Hahn, S. B. Hansen, P. Knapp, D. C. Lamppa, W. E. Lewis, L. Lucero, M. Mangan, R. Paguio, L. Perea, G. Robertson, C. Ruiz, D. Ruiz, P. F. Schmit, S. A. Slutz, G. Smith, I. Smith, C. Spears, T. Webb, M. R. Weis, K. Whittemore, E. P. Yu, R. D. McBride, K. J. Peterson, B. M. Jones, G. Rochau, and D. Sinars, “Controlling morphology and improving reproducibility of magnetized liner inertial fusion experiments,” *Phys. Plasmas* **31**, 022703 (2024).
 - ²⁶K. J. Peterson, T. J. Awe, E. P. Yu, D. B. Sinars, E. S. Field, M. E. Cuneo, M. C. Herrmann, M. Savage, D. Schroen, K. Tomlinson, and C. Nakhleh, “Electrothermal instability mitigation by using thick dielectric coatings on magnetically imploded conductors,” *Phys. Rev. Lett.* **112**, 135002 (2014).
 - ²⁷M. R. Gomez, S. A. Slutz, C. A. Jennings, D. J. Ampleford, M. R. Weis, C. E. Myers, D. A. Yager-Elorriaga, K. D. Hahn, S. B. Hansen, E. C. Harding, A. J. Harvey-Thompson, D. C. Lamppa, M. Mangan, P. F. Knapp, T. J. Awe, G. A. Chandler, G. W. Cooper, J. R. Fein, M. Geissel, M. E. Glinsky, W. E. Lewis, C. L. Ruiz, D. E. Ruiz, M. E. Savage, P. F. Schmit, I. C. Smith, J. D. Styron, J. L. Porter, B. Jones, T. R. Mattsson, K. J. Peterson, G. A. Rochau, and D. B. Sinars, “Performance scaling in magnetized liner inertial fusion experiments,” *Phys. Rev. Lett.* **125**, 155002 (2020).
 - ²⁸J. R. Fein, E. C. Harding, W. E. Lewis, M. R. Weis, and M.-A. Schaeuble, “Three-dimensional reconstruction of x-ray emission volumes in magnetized liner inertial fusion from sparse projection data using a learned basis,” *J. Appl. Phys.* **135**, 133301 (2024).
 - ²⁹R. D. McBride, M. R. Martin, R. W. Lemke, J. B. Greenly, C. A. Jennings, D. C. Rovang, D. B. Sinars, M. E. Cuneo, M. C. Herrmann, S. A. Slutz, C. W. Nakhleh, D. D. Ryutov, J.-P. Davis, D. G. Flicker, B. E. Blue, K. Tomlinson, D. Schroen, R. M. Stamm, G. E. Smith, J. K. Moore, T. J. Rogers, G. K. Robertson, R. J. Kamm, I. C. Smith, M. Savage, W. A. Stygar, G. A. Rochau, M. Jones, M. R. Lopez, J. L. Porter, and M. K. Matzen, “Beryllium liner implosion experiments on the Z accelerator in for magnetized liner inertial fusion,” *Phys. Plasmas* **20**(5), 056309 (2013).
 - ³⁰G. A. Shipley, D. E. Ruiz, and A. J. Porwitzky, “Analysis of fusion fuel conditions and instability development in shocklessly compressed MagLIF implosion simulations,” *Phys. Plasmas* **31**, 022701 (2024).
 - ³¹A. Porwitzky and J. Brown, “Accounting for slotted cylindrical anodes in current inferences on pulsed power drivers,” in *2018 IEEE International Power Modulator and High Voltage Conference (IPMHVC)* (IEEE, 2018), pp. 128–132.
 - ³²R. W. Lemke, M. D. Knudson, A. C. Robinson, T. A. Haill, K. W. Struve, J. R. Asay, and T. A. Mehlhorn, “Self-consistent, two-dimensional, magnetohydrodynamic simulations of magnetically driven flyer plates,” *Phys. Plasmas* **10**, 1867–1874 (2003).
 - ³³E. Harding, private communication (2024).
 - ³⁴W. E. Lewis, P. F. Knapp, E. C. Harding, and K. Beckwith, “Statistical characterization of experimental magnetized liner inertial fusion stagnation images using deep-learning-based fuel-background segmentation,” *J. Plasma Phys.* **88**(5), 895880501 (2022).
 - ³⁵W. von der Linden, V. Dose, and U. von Toussaint, *Bayesian Probability Theory: Applications in the Physical Sciences*, 1st ed. (Cambridge University Press, 2014).
 - ³⁶This represents the equivalent of a 2σ interval for normally distributed data.
 - ³⁷D. E. Ruiz, P. F. Schmit, D. A. Yager-Elorriaga, C. A. Jennings, and K. Beckwith, “Exploring the parameter space of MagLIF implosions using similarity scaling. I. Theoretical framework,” *Phys. Plasmas* **30**, 032707 (2023).
 - ³⁸Y. Y. Lau, J. C. Zier, I. M. Rittersdorf, M. R. Weis, and R. M. Gilgenbach, “Anisotropy and feedthrough in magneto-Rayleigh–Taylor instability,” *Phys. Rev. E* **83**, 066405 (2011).

- ³⁹N. Bennett, D. R. Welch, C. A. Jennings, E. Yu, M. H. Hess, B. T. Hutsel, G. Laity, J. K. Moore, D. V. Rose, K. Peterson, and M. E. Cuneo, "Current transport and loss mechanisms in the z accelerator," *Phys. Rev. Accel. Beams* **22**, 120401 (2019).
- ⁴⁰N. Bennett, D. R. Welch, G. Laity, D. V. Rose, and M. E. Cuneo, "Magnetized particle transport in multi-ma accelerators," *Phys. Rev. Accel. Beams* **24**, 060401 (2021).
- ⁴¹N. Bennett, D. R. Welch, K. Cochrane, K. Leung, C. Thoma, M. E. Cuneo, and G. Frye-Mason, "Electrode plasma formation and melt in z-pinch accelerators," *Phys. Rev. Accel. Beams* **26**, 040401 (2023).
- ⁴²D. D. Ryutov, T. J. Awe, S. B. Hansen, R. D. McBride, K. J. Peterson, D. B. Sinars, and S. A. Slutz, "Effect of axial magnetic flux compression on the magnetic Rayleigh-Taylor instability (theory)," *AIP Conf. Proc.* **1639**, 63–66 (2014).
- ⁴³C. E. Seyler, M. R. Martin, and N. D. Hamlin, "Helical instability in MagLIF due to axial flux compression by low-density plasma," *Phys. Plasmas* **25**, 062711 (2018).
- ⁴⁴J. M. Woolstrum, C. E. Seyler, and R. D. McBride, "Hall instability driven seeding of helical magneto-Rayleigh-Taylor instabilities in axially premagnetized thin-foil liner Z-pinch implosions," *Phys. Plasmas* **29**, 122701 (2022).
- ⁴⁵J. M. Woolstrum, D. E. Ruiz, N. D. Hamlin, K. Beckwith, and M. R. Martin, "Hall interchange instability as a seed for helical magneto-Rayleigh-Taylor instabilities in magnetized liner inertial fusion Z-Pinches scaled from Z-machine parameters to a next generation pulsed power facility," *Phys. Plasmas* **30**, 072712 (2023).
- ⁴⁶S. A. Chaikovsky, A. Yu. Labetsky, V. I. Oreshkin, A. V. Shishlov, R. B. Baksht, A. V. Fedunin, and A. G. Roussikh, "The k-shell radiation of a double gas puff z-pinch with an axial magnetic field," *Laser Part. Beams* **21**(2), 255–264 (2003).
- ⁴⁷N. Qi, P. de Grouchy, P. C. Schrafel, L. Atayan, W. M. Potter, A. D. Cahill, P.-A. Gourdain, J. B. Greenly, D. A. Hammer, C. L. Hoyt, B. R. Kusse, S. A. Pikuz, and T. A. Shelkovenko, "Gas puff Z-pinch implosions with external B_z field on COBRA," *AIP Conf. Proc.* **1639**, 51–54 (2014).
- ⁴⁸F. Conti, N. Aybar, J. Narkis, J. C. Valenzuela, H. U. Rahman, E. Ruskov, E. Dutra, S. Haque, A. Covington, and F. N. Beg, "Study of stability in a liner-on-target gas puff Z-pinch as a function of pre-embedded axial magnetic field," *Phys. Plasmas* **27**, 012702 (2020).
- ⁴⁹M. R. Weis, P. Zhang, Y. Y. Lau, P. F. Schmit, K. J. Peterson, M. Hess, and R. M. Gilgenbach, "Coupling of sausage, kink, and magneto-Rayleigh-Taylor instabilities in a cylindrical liner," *Phys. Plasmas* **22**, 032706 (2015).
- ⁵⁰A. L. Velikovich and P. F. Schmit, "Bell-Plesset effects in Rayleigh-Taylor instability of finite-thickness spherical and cylindrical shells," *Phys. Plasmas* **22**, 122711 (2015).
- ⁵¹R. Erdélyi and R. J. Morton, "Magnetohydrodynamic waves in a compressible magnetic flux tube with elliptical cross-section," *Astron. Astrophys.* **494**, 295–309 (2009).
- ⁵²D. A. Yager-Elorriaga, F. W. Doss, G. A. Shipley, P. F. Knapp, D. E. Ruiz, A. J. Porwitzky, J. R. Fein, E. C. Merritt, M. R. Martin, C. E. Myers, C. A. Jennings, I. C. Smith, D. J. Marshall, C. R. Aragon, L. Shulenburg, T. R. Mattsson, and D. B. Sinars, "Studying the Richtmyer-Meshkov instability in convergent geometry under high energy density conditions using the Decel platform," *Phys. Plasmas* **29**, 052114 (2022).
- ⁵³L. A. McPherson, D. J. Ampleford, C. A. Coverdale, J. W. Argo, A. C. Owen, and D. M. Jaramillo, "High energy x-ray pinhole imaging at the Z facility," *Rev. Sci. Instrum.* **87**, 063502 (2016).
- ⁵⁴E. C. Harding, T. Ao, J. E. Bailey, G. Loisel, D. B. Sinars, M. Geissel, G. A. Rochau, and I. C. Smith, "Analysis and implementation of a space resolving spherical crystal spectrometer for x-ray Thomson scattering experiments," *Rev. Sci. Instrum.* **86**, 043504 (2015).
- ⁵⁵D. N. Fittinghoff, D. E. Bower, J. R. Hollaway, B. A. Jacoby, P. B. Weiss, R. A. Buckles, T. J. Sammons, J. McPherson, A. Leroy, C. L. Ruiz, G. A. Chandler, J. A. Torres, R. J. Leeper, G. W. Cooper, and A. J. Nelson, "One-dimensional neutron imager for the Sandia Z facility," *Rev. Sci. Instrum.* **79**, 10E530 (2008).
- ⁵⁶D. J. Ampleford, C. L. Ruiz, D. N. Fittinghoff, J. D. Vaughan, K. Hahn, B. Lahmann, M. Gatu-Johnson, J. Frenje, R. Petrasso, C. R. Ball, A. J. Maurer, P. F. Knapp, A. J. Harvey-Thompson, J. Fisher, P. Alberto, J. A. Torres, G. Cooper, B. Jones, G. A. Rochau, and M. J. May, "One dimensional imager of neutrons on the Z machine," *Rev. Sci. Instrum.* **89**, 10I132 (2018).
- ⁵⁷S. A. Ricketts, M. A. Mangan, P. Volegov, D. N. Fittinghoff, W. E. Lewis, O. M. Mannion, J. E. Morel, M. L. Adams, and D. J. Ampleford, "Neutron source reconstruction using a generalized expectation-maximization algorithm on one-dimensional neutron images from the Z facility," *Rev. Sci. Instrum.* **95**, 033501 (2024).
- ⁵⁸C. L. Ruiz, D. L. Fehl, G. A. Chandler, G. Cooper, B. Jones, J. D. Styron, and J. Torres, "Multichannel, triaxial, neutron time-of-flight diagnostic for experiments at the Z facility," *Phys. Rev. Accel. Beams* **23**, 020401 (2020).
- ⁵⁹G. A. Chandler, C. L. Ruiz, G. W. Cooper, J. A. Torres, M. A. Mangan, G. M. Whitlow, D. J. Ampleford, M. C. Jones, R. A. Buckles, K. J. Moy, I. Garza, M. Staska, A. Wolverson, and B. Davis, "Neutron time-of-flight detectors (nTOF) used at Sandia's Z-machine," *Rev. Sci. Instrum.* **93**(11), 113531 (2022).
- ⁶⁰H. Sze, P. L. Coleman, J. Banister, B. H. Failor, A. Fisher, J. S. Levine, Y. Song, E. M. Waisman, J. P. Apruzese, R. W. Clark, J. Davis, D. Mosher, J. W. Thornhill, A. L. Velikovich, B. V. Weber, C. A. Coverdale, C. Deeney, T. L. Gilliland, J. McGurn, R. B. Spielman, K. W. Struve, W. A. Stygar, and D. Bell, "Efficient argon K-shell radiation from a Z pinch at currents > 15 MA," *Phys. Plasmas* **8**, 3135–3138 (2001).
- ⁶¹B. Jones, J. P. Apruzese, A. J. Harvey-Thompson, D. J. Ampleford, C. A. Jennings, S. B. Hansen, N. W. Moore, D. C. Lamppa, D. Johnson, M. C. Jones, E. M. Waisman, C. A. Coverdale, M. E. Cuneo, G. A. Rochau, J. L. Giuliani, J. W. Thornhill, N. D. Ouart, Y. K. Chong, A. L. Velikovich, A. Dasgupta, M. Krishnan, and P. L. Coleman, "The effect of gradients at stagnation on K-shell x-ray line emission in high-current Ar gas-puff implosions," *Phys. Plasmas* **22**, 020706 (2015).
- ⁶²A. J. Harvey-Thompson, C. A. Jennings, B. Jones, J. P. Apruzese, D. J. Ampleford, D. C. Lamppa, C. A. Coverdale, M. E. Cuneo, J. L. Giuliani, S. B. Hansen, M. C. Jones, N. W. Moore, G. A. Rochau, and J. W. Thornhill, "Investigating the effect of adding an on-axis jet to Ar gas puff Z pinches on Z," *Phys. Plasmas* **23**, 101203 (2016).